

Geethanjali College of Engineering and Technology

AUTONOMOUS

(Accredited by NAAC "A+" Grade ECE,CSE,CIVIL,EEE & ME, B.Tech Program Accredited by NBA;
Approved by AICTE, New Delhi, Affiliated to JNTUH)

Cheeryal (V), Keesara (M), Medchal (Dist), Telangana state.



STATISTICS FOR MACHINE LEARNING LABORATORY WORKBOOK

III Year B.Tech -I Semester
(2023-2024)

Name: Roll No:

Branch: Section:

Geethanjali College of Engineering and Technology

AUTONOMOUS

(Accredited by NBA, NAAC “A+” Grade)

Cheeryal (V), Keesara (M), Medchal District – 501 301 (T S)



STATISTICS FOR MACHINE LEARNING LAB

Subject Code – (20MA31L01)

INSTRUCTOR LABAORATORY MANUAL

III Year B.Tech I Semester

Common to

ELECTRICAL AND ELECTRONICS ENGINEERING – (EEE),

MECHANICAL ENGINEERING - (ME)

COMPUTER SCIENCE AND ENGINEERING - ARTIFICIAL INTELLIGENCE AND MACHINE
LEARNING - CSE(AIML),

COMPUTER SCIENCE AND ENGINEERING -DATA SCIENCE - CSE(DS),

COMPUTER SCIENCE AND ENGINEERING - CYBER SECURITY - CSE(CS)

2023-2024

DEPARTMENT OF FRESHMAN ENGINEERING

Lab Incharge

HoD

Geethanjali College of Engineering and Technology DEPARTMENT OF FRESHMAN ENGINEERING (Name of the Lab Course): STATISTICS FOR MACHINE LEARNING LAB (Subject Code): 20MA31L01 Programme : UG			
Branch : EEE, ME, CSE_AIML, CSE_DS, CSE_CS Year : III Semester : I	Version No : 1 Updated on : 14/08/2023 No. of pages : 75		
Classification status (Unrestricted / Restricted) Distribution List: Department, Lab, Library, Lab In Charge			
<table style="width: 100%; border: none;"> <tr> <td style="width: 50%; vertical-align: top;"> Prepared by: Dr T. Deepthi 1) Name : Dr T. Deepthi 2) Sign : 3) Design : Associate Professor 4) Date : 14.08.2023 </td> <td style="width: 50%; vertical-align: top;"> Modified by: 1) Name : 2) Sign : 3) Design : Asst. Prof 4) Date : </td> </tr> </table>		Prepared by: Dr T. Deepthi 1) Name : Dr T. Deepthi 2) Sign : 3) Design : Associate Professor 4) Date : 14.08.2023	Modified by: 1) Name : 2) Sign : 3) Design : Asst. Prof 4) Date :
Prepared by: Dr T. Deepthi 1) Name : Dr T. Deepthi 2) Sign : 3) Design : Associate Professor 4) Date : 14.08.2023	Modified by: 1) Name : 2) Sign : 3) Design : Asst. Prof 4) Date :		
Verified by : 1) Name : 2) Sign : 3) Design : Professor 4) Date :	* <u>For Q.C Only.</u> 1) Name : 2) Sign : 3) Design : 4) Date :		
Approved by : (HOD) 1) Name : 2) Sign : 3) Date			

Geethanjali College of Engineering and Technology

Geethanjali College of Engineering and Technology
AUTONOMOUS

Subject code: 20MA31L01

Pre-requisite(s): 20MA31001- Statistics for Machine Learning

L	T	P/D	C
-	-	2/-	1

Course Outcomes:

At the end of the course, student would be able to use R software to:

CO1: Construct frequency distribution for the given data.

CO2: Obtain specified set of samples from large data through random variables using Acceptance-Rejection (AR) algorithm.

CO3: Estimate and test the significance of the specified population parameter(s) for the given data.

CO4: Reduce the dimensionality of large data sets using Principal Component Analysis (PCA), Logistic Regression (LR), K-means clustering and Support Vector Machine (SVM) for selecting variables.

List of Experiments:

1. Introduction to R Programming
2. Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)
3. Acceptance/Rejection Sampling in R
4. Maximum likelihood and method of moments estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R
5. Testing of hypothesis based on t-test, chi-square test and F-test.
Confidence interval estimation in R
6. Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R
7. Logistic Regression in R
8. Classification with SVM in R
9. Dimensionality reduction with PCA in R and K-means Clustering in R
10. Kernel PCA and Kernel SVM in R

List of Additional programs

1. Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R programming
2. Multiple linear regression: Testing overall hypothesis and testing significance of individual variables, model selections and prediction in R programming.

Vision of the Institute:

Geethanjali visualizes dissemination of knowledge and skills to students, who would eventually contribute to the well-being of the people of the nation and global community.

Mission of the Institute:

- To impart adequate fundamental knowledge in all basic sciences and engineering, technical and Interpersonal skills to students.
- To bring out creativity in students that would promote innovation, research and entrepreneurship.
- To Preserve and promote cultural heritage, humanistic and spiritual values promoting peace and harmony in society.

Vision of the Department:**Department of EEE**

To provide excellent Electrical and electronics education by building strong teaching and research environment.

Department of ME

The Mechanical Engineering department strives to be recognized globally for outstanding education and research, imparting quality education, churning well-qualified engineers, who are creative, innovative, and entrepreneurial, solving problems for societal development.

Department of CSE-AIML, CSE-DS, CSE-CS

To produce globally competent and socially responsible computer science engineers contributing to the advancement of engineering and technology which involves creativity and innovation by providing excellent learning environment with world class facilities.

Mission of the Department:**Department of EEE**

- To offer high quality graduate program in Electrical and Electronics education and to prepare students for professional career or higher studies.
- The department promotes excellence in teaching, research, collaborative activities and positive contributions to society.

Department of ME

- Imparting quality education to students to enhance their skills and make them globally competitive.
- Prepare graduates to engage in life-long learning, possess intellectual capabilities, serving society with a strong commitment to their profession, meeting technical challenges and exhibiting ethical responsibility for societal development.
- Conduct quality research, create opportunities for students and faculty to showcase their talent, disseminate knowledge, and promote community development, leading to peace and harmony in society.

Department of CSE-AIML, CSE-DS, CSE-CS

1. To be a centre of excellence in instruction, innovation in research and scholarship, and service to the stake holders, the profession, and the public.

2. To prepare graduates to enter a rapidly changing field as a competent computer science engineer.
3. To prepare graduate capable in all phases of software development, possess a firm understanding of hardware technologies, have the strong mathematical background necessary for scientific computing, and be sufficiently well versed in general theory to allow growth within the discipline as it advances.
4. To prepare graduates to assume leadership roles by possessing good communication skills, the ability to work effectively as team members, and an appreciation for their social and ethical responsibility in a global setting

Program Educational Objectives (PEO's):

EEE

- PEO 1:** To prepare students with excellent comprehension of mathematics, basic sciences and engineering subjects facilitating them to find gainful employment or pursue postgraduate program with an appreciation for lifelong learning.
- PEO 2:** To inculcate problem solving capabilities in students with analysis, design and practical skills that are Program Specific which would facilitate them to exhibit creativity and innovation that would enable them to develop modern equipment with emerging technologies of multidisciplinary nature for societal development.
- PEO 3:** To inculcate positive attitude, professional ethics, effective communication and interpersonal skills which would facilitate them to succeed in the chosen profession through research and development both as team member and as well as leader.

ME

- PEO 1:** To prepare students with strong fundamental knowledge in basic sciences mathematics, engineering courses which would facilitate them find gainful employment with a sense of appreciation to pursue life-long learning for professional development.
- PEO 2:** To inculcate problem solving skills in students, imbibing creativity and innovation which would enable them to develop modern machinery involving cutting edge technologies of Multidisciplinary nature for societal development.
- PEO 3:** To develop critical thinking with an aptitude to conduct research and development, instill professional ethics, develop effective communication and interpersonal skills with positive attitude to contribute significantly towards their chosen profession, thereby supporting community development.

CSE (AI&ML, DS, CS)

- PEO 1:** To provide graduates with a good foundation in mathematics, sciences and engineering fundamentals required to solve engineering problems that will facilitate them to find employment in industry and / or to pursue postgraduate studies with an appreciation for lifelong learning.
- PEO 2:** To provide graduates with analytical and problem-solving skills to design algorithms, other hardware / software systems, and inculcate professional ethics, inter-personal skills to work in a multi-cultural team.

PEO 3: To facilitate graduates to get familiarized with the art software / hardware tools, imbibing creativity and innovation that would enable them to develop cutting-edge technologies of multi-disciplinary nature for societal development.

Program Outcomes (POs):

- 1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- 2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- 3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- 4. Conduct investigations of complex problems:** The problems
 - that cannot be solved by straightforward application of knowledge, theories and techniques applicable to the engineering discipline.
 - that may not have a unique solution. For example, a design problem can be solved in many ways and lead to multiple possible solutions.
 - that require consideration of appropriate constraints/requirements not explicitly given in the problem statement. (Like: cost, power requirement, durability, product life, etc.).
 - which need to be defined (modeled) within appropriate mathematical framework.
 - that often require use of modern computational concepts and tools.
- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- 6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- 7. Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- 8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- 9. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- 10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- 11. Project management and finance:** Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- 12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs):

EEE

- PSO1:** Ability to apply fundamental knowledge to identify, interpret, formulate, design, analyze and investigate various problems of electrical and electronic systems/components and integrate them into generation, transmission, distribution and utilization of electrical energy through conventional and non-conventional energy sources.
- PSO2:** Ability to apply emerging technologies in the design, simulation and analysis of electrical and electronic systems and demonstrate capabilities of scientific and innovative thinking to meet the technical challenges of the society and industry.

ME

- PSO 1:** Apply Continuity, Energy and Momentum equations to mechanical systems, design and perform experiments in all fields of mechanical engineering
- PSO 2:** Able to analyze, design and develop/model mechanical and its allied systems using software tools such as AUTOCAD, ANSYS, Creo etc.
- PSO 3:** Able to design layouts for process and manufacturing industry taking into consideration optimization of resources for effective operation and maintenance.

(CSE-AI&ML)

- PSO 1:** Demonstrate competency in Programming and problem-solving skills and apply those skills in solving computing problems.
- PSO 2:** Select appropriate programming languages, Data structures and algorithms in combination with modern technologies and apply them in developing innovative Solutions.
- PSO 3:** Demonstrate adequate knowledge in the concepts and techniques of artificial intelligence and machine learning, apply them in developing intelligent systems to solve real world problems.

(CSE-DS)

- PSO 1:** Demonstrate competency in Programming and problem-solving skills and apply those skills in solving computing problems.
- PSO2:** Select appropriate programming languages, Data structures and algorithms in combination with modern technologies and apply them in developing innovative solutions.
- PSO 3:** Apply techniques of data modeling, analysis and visualization which include statistical techniques to solve real world problems delivering actionable insights for decision making.

(CSE-CS)

- PSO 1:** Demonstrate competency in Programming and problem-solving skills and apply those skills in solving computing problems.
- PSO2:** Select appropriate programming languages, Data structures and algorithms in combination with modern technologies and apply them in developing innovative solutions.

PSO3: Apply cryptography algorithms for ensuring cyber security as per cyber laws, demonstrate awareness of all the security related issues and employ state of the art technologies to protect the digital assets of an organization.

Mapping of Lab Course outcomes with Program outcomes:

EEE

CO-PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	2	2	-	-	-	-	-	-	-	-	2	-	-	-
CO2	3	2	2	-	-	-	-	-	-	-	-	2	-	-	-
CO3	3	2	2	-	3	-	-	-	-	-	-	3	-	2	-
CO4	3	2	2	-	3	-	-	-	-	-	-	3	-	2	-

-	No Correlation	1	Slight (low)	2	Moderate (medium)	3	Substantial (High)
---	----------------	---	--------------	---	-------------------	---	--------------------

ME

CO-PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	2	2	-	-	-	-	-	-	-	-	2	-	-	-
CO2	3	2	2	-	-	-	-	-	-	-	-	2	-	-	-
CO3	3	2	2	-	3	-	-	-	-	-	-	3	-	2	-
CO4	3	2	2	-	3	-	-	-	-	-	-	3	-	2	-

-	No Correlation	1	Slight (low)	2	Moderate (medium)	3	Substantial (High)
---	----------------	---	--------------	---	-------------------	---	--------------------

CSE-AIML

CO-PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	2	2	-	-	-	-	-	-	-	2	2	-	-	-
CO2	3	2	2	-	-	-	-	-	-	-	2	2	-	-	-
CO3	3	2	-	-	3	-	-	-	-	-	2	3	2	-	3
CO4	3	2	-	-	3	-	-	-	-	-	2	3	2	-	3

-	No Correlation	1	Slight (low)	2	Moderate (medium)	3	Substantial (High)
---	----------------	---	--------------	---	-------------------	---	--------------------

CSE-DS

CO-PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	2	2	-	-	-	-	-	-	-	2	2	-	-	-
CO2	3	2	2	-	-	-	-	-	-	-	2	2	-	-	-
CO3	3	2	-	-	3	-	-	-	-	-	2	3	2	-	3
CO4	3	2	-	-	3	-	-	-	-	-	2	3	2	-	3

-	No Correlation	1	Slight (low)	2	Moderate (medium)	3	Substantial (High)
---	----------------	---	--------------	---	-------------------	---	--------------------

CSE-CS

CO-PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	2	2	-	-	-	-	-	-	-	2	2	-	-	-
CO2	3	2	2	-	-	-	-	-	-	-	2	2	-	-	-
CO3	3	2	-	-	3	-	-	-	-	-	2	3	2	-	-
CO4	3	2	-	-	3	-	-	-	-	-	2	3	2	-	-

-	No Correlation	1	Slight (low)	2	Moderate (medium)	3	Substantial (High)
---	----------------	---	--------------	---	-------------------	---	--------------------

Prerequisites:

Statistics for Machine Learning (20MA31001)

INSTRUCTIONS TO THE STUDENTS:

1. Students are required to attend all labs.
2. Students should be dressed in formals when attending the laboratory sessions.
3. Students will work individually in computer laboratories.
4. While coming to the lab bring the observation book and Work book etc.
5. Before coming to the lab, prepare the pre-lab questions. Read through the lab experiment to familiarize you.
6. Utilize 3 hours' time properly to perform the experiment and noting down the outputs.
7. If the experiment is not completed in the prescribed time, the pending work has to be done in the leisure hour or extended hours.
8. You will be expected to submit the completed work book according to the deadlines set up by your instructor.

INSTRUCTIONS TO LABORATORY TEACHERS:

1. Observation book and lab records submitted for the lab work are to be checked and signed before the next lab session.

2. Students should be instructed to switch ON the power supply after the connections are checked by the lab assistant / teacher.
3. The promptness of submission should be strictly insisted by awarding the marks accordingly.
4. Ask Viva Voce questions at the end of the experiment.
5. Do not allow students who come late to the lab class.
6. Encourage the students to do the experiments innovatively.
7. Fill Continuous Evaluation Sheet, on regular basis.
8. Ensure that the students are dressed in formals

Evaluation pattern:

Total Lab Marks –100 (Internal - 30 Marks & External - 70 Marks)

Internal Assessment:

1. Continuous evaluation is done for 15 Marks based on:
 - a. Performance of Program (5 Marks)
 - b. Submission of workbook in time (5 Marks)
 - c. Viva-Voce (5 Marks)
2. During the semester two Lab Internal examinations are conducted for 15 Marks and average is taken.

External Assessment:

1. At the end of semester, external lab exam is conducted for 70 Marks with one external examiner and one internal examiner, as per regulation.
 - a. Write up and program (30 Marks)
 - b. Execution of Program (30 Marks)
 - c. Viva-Voce (10 Marks)

PERFORMANCE INDICATOR

S No.	Name of the Program	Page No.	Date of Programme	Date of Submission	Marks awarded	Faculty Sign.	Remarks
1	Introduction to R Programming						
2	Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)						
3	Acceptance/Rejection Sampling in R						
4	Maximum likelihood and method of moments estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R						
5	Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R						
6	Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R						
7	Logistic Regression in R						
8	Classification with SVM in R						
9	Dimensionality reduction with PCA in R and K-means Clustering in R						
10	Kernal SVM in R						

	ADDITIONAL PROGRAMMES						
S No.	Name of the Program	Page No.	Date of Programme	Date of Submission	Marks awarded	Faculty Sign.	Remarks
1	Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R programming						
2	Multiple linear regression: Testing over all hypothesis and testing significance of individual variables, model selections and prediction in R programming						

S No.	INDEX	Page No.
1	Introduction to R Programming	14
2	Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)	19
3	Acceptance/Rejection Sampling in R	26
4	Maximum likelihood and method of moments estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R	31
5	Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R	39
6	Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R	44
7	Logistic Regression in R	50
8	Classification with SVM in R	53
9	Dimensionality reduction with PCA in R and K-means Clustering in R	58
10	Kernal PCA and Kernal SVM in R	66
	Additional Programmes	
1	Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R programming	71
2	Multiple linear regression: Testing overall hypothesis and testing significance of individual variables, model selections and prediction in R programming	73

1. Introduction to R Programming

R is a programming language and free software environment for statistical computing and graphics that is supported by the R Foundation for Statistical Computing. The R language is widely used among statisticians and data miners for developing statistical software and data analysis.

R is a programming language and software environment for statistical analysis, graphics representation and reporting.

The following are the important features of R

- R is a well-developed, simple and effective programming language which includes conditionals, loops; user defined recursive functions and input and output facilities.
- R has an effective data handling and storage facility,
- R provides a suite of operators for calculations on arrays, lists, vectors and matrices.
- R provides a large, coherent and integrated collection of tools for data analysis.
- R provides graphical facilities for data analysis and display either directly at the computer or printing at the papers.

To Install R and R Packages:

1. Open an internet browser and go to www.r-project.org.
2. Click the "download R" link in the middle of the page under "Getting Started."
3. Select a CRAN location (a mirror site) and click the corresponding link.
4. Click on the "Download R for WINDOWS" link at the top of the page.
5. Click on the file containing the latest version of R under "Files."
6. Save the **.pkg** file, double-click it to open, and follow the installation instructions.
7. Now that R is installed, you need to download and install R Studio.

To Install R Studio:

1. Go to www.rstudio.com and click on the "Download R Studio" button.
2. Click on "Download R Studio Desktop."
3. Click on the version recommended for your system, or the latest Mac version, save the **.dmg** file on your computer, double-click it to open, and then drag and drop it to your applications folder.

Install R Packages:

1. Download and install a package (you only need to do this once).
2. To use the package, invoke the library (package) command to load it into the current session. (You need to do this once in each session, unless you customize your environment to automatically load it each time).

Installing and Loading Packages:

R to load the package's library in R session

```
> library ("MASS") #R now knows all the functions that are scanned in the MASS package
```

1.1. Basics and R as a Calculator:

i. $31 * 78$ # Multiplication

```
[1] 2418
```

ii. $697/41$ #division

```
[1] 17.9
```

iii. Assign the value of 39 to x

```
> x<-39
```

```
> x
```

```
[1] 39
```

iv. Assign the value of 22 to y

```
> y<-22
```

```
> y
```

```
[1] 22
```

v. Make z the value of $x - y$

```
> z<- x - y
```

vi. Display the value of z in the console

```
> z
```

```
[1] 17
```

vii. Find 10 random numbers between 1 and 100. (Hint: you can use sample() function)

```
> sample(1:100,10)
```

1 Example: Find the value of 21 is multiplied by 23, 32 is multiplied by 62; 1256 divided by 2; Assign the value of 26 to x variable, 33 to y variable and hence find the value of z such that $z = x+y$; $x-y$;

R Code with Output:

```
> 21*23
```

```
[1] 483
```

```
> 32*62
```

```
[1] 1984
```

```
> 1256/2
```

```
[1] 628
```

```
> x=26
```

```
> y=33
```

```
> z=x+y
> z
[1] 59
> x-y
[1] -7
```

2 Example: Find 5 random numbers between 1 to 100 using sample function

R Code with Output:

```
> sample(1:100,5)
[1] 23 94 30 62 45
```

1.2. Creating Vectors:

A vector is a basic data structure of R containing the same type of data.

R Code

x <- c(observations / data)

1. **Example:** The numbers below are the first ten days of rainfall amounts in 1996. Read them into a vector using the vector function named as “rainfall”

0.1, 0.6, 33.8, 1.9, 9.6, 4.3, 33.7, 0.3, 0.0, 0.1

R-Code with Output:

```
> rainfall=c(0.1,0.6,33.8,1.9,9.6,4.3,33.7,0.3,0.0,0.1)
> rainfall
[1] 0.1 0.6 33.8 1.9 9.6 4.3 33.7 0.3 0.0 0.1
```

1.3. Creating Matrices:

To create matrices we will use the matrix() function. The matrix() function takes the following arguments:

- data an R object (this could be a vector).
- nrow the desired number of rows.
- ncol the desired number of columns.
- by row a logical statement to populate the matrix by either row or by column

R Code

matrix1 <- matrix (data = 1, nrow = 3, ncol = 3)

> matrix1

Output:

```

      [,1] [,2] [,3]
[1,]    1    1    1
[2,]    1    1    1
[3,]    1    1    1

```

1. Example: If x is a matrix specified by the R command `x = matrix(nrow=3, ncol=3, data=c(1,8,2,1,5,4,3,9,6), byrow=TRUE)`, then the output of the R command `x%*%t(x)` is?

R-Code with Output:

```
> x=matrix(nrow=3,ncol=3,data=c(1,8,2,1,5,4,3,9,6), byrow =TRUE)
```

```
> x%*%t(x)
```

```

      [,1] [,2] [,3]
[1,]  69  49  87
[2,]  49  42  72
[3,]  87  72 126

```

2. Example: If $A = \begin{bmatrix} 0 & 3 & 1 \\ 5 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ are two matrices then check whether the two matrices satisfy commutative property or not. And also find transpose of each matrix using R program.

R-Code with Output:

```

> A=matrix(nrow=3,ncol=3,data=c(0,3,1,5,1,-1,1,2,1), byrow =TRUE)
> B=matrix(nrow=3,ncol=3,data=c(0,0,1,0,1,0,1,0,1), byrow =TRUE)
> A%*%B
      [,1] [,2] [,3]
[1,]    1    3    1
[2,]   -1    1    4
[3,]    1    2    2
> B%*%A
      [,1] [,2] [,3]
[1,]    1    2    1
[2,]    5    1   -1
[3,]    1    5    2

```

From the above matrices $A*B \neq B*A$. Hence two matrices A & B are not commutative.

```

> t(A)
      [,1] [,2] [,3]
[1,]    0    5    1
[2,]    3    1    2
[3,]    1   -1    1
> t(B)
      [,1] [,2] [,3]
[1,]    0    0    1
[2,]    0    1    0
[3,]    1    0    1

```

Viva Questions:

1. Write the syntax for installing package MASS?

Ans. `install.packages("MASS")`

2. Write the syntax to load MASS package in library?

Ans. `library(MASS)`

3. Write the R code to find the value of $Z = X + Y$. where $X = 26$, $Y = -10$.

Ans. `Z = 16`

2. Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)

Absolute Frequency, Relative Frequency:

The number of observations in a particular class / category is called the **Absolute Frequency** or **Frequency**.

The number of observations in a particular class / category is divided by total frequency is called **Relative Frequency**.

2.1. Absolute and Relative Frequency.

R Code:

table(data vector) #creates the *absolute frequency* of the **data vector** of the given data

Enter data as x

table(x) # displays Absolute Frequencies

table(x)/length(x) # displays Relative Frequencies

1. Example: Results of 10 persons declared in two categories as Pass (P) and Fail (F) is categorized as 1 and 2 respectively. Find Absolute and Relative Frequencies.

P, F, P, F, F, P, P, F, P, P
1, 2, 1, 2, 2, 1, 1, 2, 1, 1

R Code with Output:

```
> x=c(1,2,1,2,2,1,1,2,1,1)
```

```
> table(x)                > table(x)/length(x)
```

x	x
1 2	1 2
6 4	0.6 0.4

2. Example: The time (in minutes) taken to complete the puzzle was recorded as follows and stored in a data vector as puzzle time. Find Absolute and Relative frequencies.

8,13,21,9,15,9,21,13,8,8,13,15,8,21,9,15,8,13,21

R Code with Output:

```
> x=c(8,13,21,9,15,9,21,13,8,8,13,15,8,21,9,15,8,13,21)
```

```
> table(x)
```

```
x
8 9 13 15 21
5 3 4 3 4
```

```
> table(x)/length(x)
```

x	8	9	13	15	21
	0.2631579	0.1578947	0.2105263	0.1578947	0.2105263

2.2. Frequency Distribution: Arrangement of ungrouped data in the form of group is called

Frequency Distribution of data.

Algorithm:

- First step is to find the “range” of the data values which can be partitioned into class interval.
Use command `range` which returns a vector containing the minimum and maximum of all the given arguments.
- Create a sequence starting from lower limit to upper limit of given set of observations at an interval of width specified in the given problem.
- Now we need to convert Numeric to Factor using a command **cut**
- Now we can compute the absolute frequency of given data in each width interval with the **table()**
- Use the **cbind** function to print the frequency distribution in column format.
- To compute the relative frequency distribution of given data in each width interval with **table()** by **length()**.

2.3. Cumulative Distribution Function (CDF) for data: It gives us an idea about the cumulative frequencies up to a certain point of the given data using **cumsum()**.

R Code:

```
>x=c(data vector)
```

```
>range(x) #returns a vector containing the minimum and maximum of all the given arguments.
```

```
>breaks = seq(minimum value, maximum value, by=specified width)
```

```
# Sequence at interval of specified width.
```

```
>C_I=cut(x, breaks, right = FALSE) #divides the range of data vector into intervals and codes the values
```

```
in data vector according to which interval they fall
```

```
>Freq=table(C_I) # gives the absolute frequency of the variable of the data file which generates the frequency distribution of the data on variable
```

```
>cbind(Freq) # print the frequency distribution in column format
```

```
>Rel_Freq=table(C_I)/length(C_I) #gives the relative frequency of the variable of the data file which
```

```
generates the frequency distribution of the data on variable
```

```
>cbind(Rel_Freq) # print the relative frequency distribution in column format.
```

R Code for Cumulative Distribution function:

>Cum_Freq=cumsum(Freq) #returns a vector whose elements are the cumulative sums of the elements

of the absolute frequency distribution.

>cbind(Cum_Freq) # print the cumulative frequency distribution in column format

Note: Suppose if we want to display the total Frequency distribution table consisting of Absolute Frequency, Relative Frequency and Cumulative Frequency in column Format then:

> cbind(Freq, Rel_Freq, Cum_Freq)

1. Example:

Following are the 50 test scores for male students in an examination.

72, 55, 35, 56, 80, 68, 52, 49, 35, 20, 30, 23, 48, 90, 24, 58, 65, 29, 50, 67, 37, 42, 36, 86, 63, 78, 84, 60, 22, 57, 61, 79, 70, 85, 64, 41, 69, 66, 28, 89, 47, 88, 72, 21, 81, 75, 32, 27, 73, 31.

a) Write R code to construct frequency distribution table of width 10.

b) And also find relative frequency distribution and cumulative frequency distribution for the male students.

c) Represent the above (a) and (b) in column format.

R code with Output:

```
>x=c(72, 55, 35, 56, 80, 68, 52, 49, 35, 20, 30, 23, 48, 90, 24, 58, 65, 29, 50, 67, 37, 42, 36, 86, 63, 78, 84, 60, 22, 57, 61, 79, 70, 85, 64, 41, 69, 66, 28, 89, 47, 88, 72, 21, 81, 75, 32, 27, 73, 31)
```

```
>range(x)
```

```
>breaks=seq(15,100,by=10)
```

```
>C_I=cut(x, breaks, right = FALSE)
```

```
  a) >Freq=table(C_I)
```

```
      >Freq
```

```
  b) >Rel_Freq=table(C_I)/length(C_I)  and  Cum_Freq=cumsum(Freq)
```

```
      >Rel_Freq
```

```
      >Cum_Freq
```

```
  c) cbind(Freq, Rel_Freq, Cum_Freq)
```

```

> range(x)
[1] 20 90
> breaks=seq(15,100,by=10)
> C_I=cut(x, breaks, right = FALSE)
> Freq=table(C_I)
> Freq
C_I
[15,25) [25,35) [35,45) [45,55) [55,65) [65,75) [75,85) [85,95)
      5      6      6      5      8      9      6      5
> Rel_Freq=table(C_I)/length(C_I)
> Rel_Freq
C_I
[15,25) [25,35) [35,45) [45,55) [55,65) [65,75) [75,85) [85,95)
  0.10  0.12  0.12  0.10  0.16  0.18  0.12  0.10
> Cum_Freq=cumsum(Freq)
> Cum_Freq
[15,25) [25,35) [35,45) [45,55) [55,65) [65,75) [75,85) [85,95)
      5     11     17     22     30     39     45     50
> cbind(Freq, Rel_Freq, Cum_Freq)
      Freq Rel_Freq Cum_Freq
[15,25)    5   0.10      5
[25,35)    6   0.12     11
[35,45)    6   0.12     17
[45,55)    5   0.10     22
[55,65)    8   0.16     30
[65,75)    9   0.18     39
[75,85)    6   0.12     45
[85,95)    5   0.10     50

```

2. Example:

Following are the scores of 250 students in a test with minimum and maximum marks being 0 and 100 respectively. The scores are stored in a data vector named **scores**.

99.86, 42.72, 3.66, 66.89, 96.60, 30.08, 49.50, 16.84, 89.14, 14.97, 66.86, 38.32, 71.90, 9.50, 66.92, 58.83, 38.06, 79.95, 95.30, 97.51, 49.67, 50.54, 93.64, 74.30, 76.82, 63.32, 36.54, 90.87, 72.37, 61.98, 89.44, 98.19, 27.07, 63.97, 55.35, 92.40, 32.97, 61.37, 34.40, 42.48, 77.67, 99.31, 54.64, 69.46, 49.88, 86.76, 31.95, 4.55, 36.77, 36.91, 56.45, 54.44, 76.53, 15.97, 18.22, 41.34, 50.17, 18.82, 96.60, 93.58, 8.85, 30.27, 44.18, 26.85, 54.82, 64.24, 46.20, 3.00, 96.14, 10.01, 75.64, 93.18, 93.32, 15.88, 84.66, 21.08, 13.76, 75.89, 68.41, 51.36, 21.58, 13.85, 82.75, 71.88, 57.93, 6.68, 24.99, 10.24, 24.60, 95.14, 37.91, 97.11, 66.32, 87.91, 85.58, 64.32, 16.11, 37.22, 42.69, 89.64, 89.96, 48.29, 27.71, 89.71, 23.08, 53.02, 4.22, 18.84, 17.41, 73.05, 33.26, 71.83, 8.28, 69.65, 51.27, 75.61, 63.67, 8.24, 1.15, 83.05, 23.08, 26.37, 19.74, 34.79, 53.58, 66.88, 99.58, 63.30, 60.82, 43.71, 39.76, 54.97, 13.55, 81.42, 26.04, 2.18, 69.18, 36.76, 67.77, 40.94, 37.88, 6.44, 24.03, 31.21, 44.02, 50.53, 1.29, 78.89, 78.18, 41.64, 13.12, 60.07, 37.38, 7.18, 38.98, 75.08, 17.65, 11.13, 19.43, 44.85, 63.36, 68.78, 91.34, 71.68, 90.47, 82.64, 15.24, 24.96, 14.07, 1.79, 35.54, 62.56, 76.10, 7.22, 96.00, 49.04, 93.72, 41.35, 6.35, 31.58, 14.91, 46.10, 48.83, 25.15, 15.98, 18.06, 54.47, 88.50, 87.06, 16.53, 47.53, 46.51, 68.91, 28.35, 68.89, 93.41, 21.07, 25.42, 93.02, 94.64, 93.94, 9.75, 89.52, 1.20, 34.41, 65.48, 58.09, 32.27, 55.53, 22.79, 5.41, 87.13, 52.83, 14.59, 17.07, 29.37, 94.04, 24.53, 62.59, 58.07,

83.71, 86.55, 23.05, 34.52, 35.22, 50.84, 71.60, 27.66, 5.29, 72.83, 34.59, 68.72, 6.51, 29.91, 40.08, 45.36, 16.20, 1.91, 22.10, 86.63, 44.83, 6.13, 93.60, 73.23, 2.01, 5.79, 72.59, 94.20, 1.20, 91.42

a) Write R code to construct frequency distribution table of width 20.

b) And also find relative frequency distribution and cumulative frequency distribution for the male students.

c) Represent the above (a) and (b) in column format.

R code with Output:

```
>x=c(99.86, 42.72, 3.66, 66.89, 96.60, 30.08, 49.50, 16.84, 89.14, 14.97, 66.86, 38.32, 71.90, 9.50,
66.92, 58.83, 38.06, 79.95, 95.30, 97.51, 49.67, 50.54, 93.64, 74.30, 76.82, 63.32, 36.54, 90.87,
72.37, 61.98, 89.44, 98.19, 27.07, 63.97, 55.35, 92.40, 32.97, 61.37, 34.40, 42.48, 77.67, 99.31,
54.64, 69.46, 49.88, 86.76, 31.95, 4.55, 36.77, 36.91, 56.45, 54.44, 76.53, 15.97, 18.22, 41.34, 50.17,
18.82, 96.60, 93.58, 8.85, 30.27, 44.18, 26.85, 54.82, 64.24, 46.20, 3.00, 96.14, 10.01, 75.64, 93.18,
93.32, 15.88, 84.66, 21.08, 13.76, 75.89, 68.41, 51.36, 21.58, 13.85, 82.75, 71.88, 57.93, 6.68, 24.99,
10.24, 24.60, 95.14, 37.91, 97.11, 66.32, 87.91, 85.58, 64.32, 16.11, 37.22, 42.69, 89.64, 89.96,
48.29, 27.71, 89.71, 23.08, 53.02, 4.22, 18.84, 17.41, 73.05, 33.26, 71.83, 8.28, 69.65, 51.27, 75.61,
63.67, 8.24, 1.15, 83.05, 23.08, 26.37, 19.74, 34.79, 53.58, 66.88, 99.58, 63.30, 60.82, 43.71, 39.76,
54.97, 13.55, 81.42, 26.04, 2.18, 69.18, 36.76, 67.77, 40.94, 37.88, 6.44, 24.03, 31.21, 44.02,
50.53, 1.29, 78.89, 78.18, 41.64, 13.12, 60.07, 37.38, 7.18, 38.98, 75.08, 17.65, 11.13, 19.43, 44.85,
63.36, 68.78, 91.34, 71.68, 90.47, 82.64, 15.24, 24.96, 14.07, 1.79, 35.54, 62.56, 76.10, 7.22, 96.00,
49.04, 93.72, 41.35, 6.35, 31.58, 14.91, 46.10, 48.83, 25.15, 15.98, 18.06, 54.47, 88.50, 87.06, 16.53,
47.53, 46.51, 68.91, 28.35, 68.89, 93.41, 21.07, 25.42, 93.02, 94.64, 93.94, 9.75, 89.52, 1.20, 34.41,
65.48, 58.09, 32.27, 55.53, 22.79, 5.41, 87.13, 52.83, 14.59, 17.07, 29.37, 94.04, 24.53, 62.59, 58.07,
83.71, 86.55, 23.05, 34.52, 35.22, 50.84, 71.60, 27.66, 5.29, 72.83, 34.59, 68.72, 6.51, 29.91, 40.08,
45.36, 16.20, 1.91, 22.10, 86.63, 44.83, 6.13, 93.60, 73.23, 2.01, 5.79, 72.59, 94.20, 1.20, 91.42)
>range(x)
>breaks=seq(1,100,by=10)
>C_I=cut(x, breaks, right = FALSE)
a)>Freq=table(C_I)
>Freq
b)>Rel_Freq=table(C_I)/length(C_I) and Cum_Freq=cumsum(Freq)
>Rel_Freq
>Cum_Freq
c)cbind(Freq, Rel_Freq, Cum_Freq)
```

```

> range(x)
[1] 1.15 99.86
> breaks=seq(0,100,by=20)
> C_I=cut(x, breaks, right = FALSE)
> Freq=table(C_I)
> Freq
C_I
[0,20) [20,40) [40,60) [60,80) [80,100)
    55     51     45     50     49
> Rel_Freq=table(C_I)/length(C_I)
> Rel_Freq
C_I
[0,20) [20,40) [40,60) [60,80) [80,100)
 0.220  0.204  0.180  0.200  0.196
> Cum_Freq=cumsum(Freq)
> Cum_Freq
[0,20) [20,40) [40,60) [60,80) [80,100)
    55    106    151    201    250
> cbind(Freq, Rel_Freq, Cum_Freq)
      Freq Rel_Freq Cum_Freq
[0,20)   55  0.220     55
[20,40)   51  0.204    106
[40,60)   45  0.180    151
[60,80)   50  0.200    201
[80,100)  49  0.196    250

```

Viva Questions

1. Write R Code to represent the values in column format.

`cbind()`

2. Define range

The difference between maximum and minimum value

3. Write R code to find the relative frequencies

`table(datavector)/length(datavector)`

3. Acceptance/Rejection Sampling in R

Algorithm

1. Find a r.v. Y with density “ g ” satisfying $f(t)/g(t) \leq c$, for all t such that $f(t) > 0$.
2. Generate a random y from the dist. with density g
3. Generate a random u from the $U(0, 1)$ dist.
4. If $u < f(y)/(cg(y))$ accept y and deliver $x = y$; otherwise reject y and repeat step 2-4.

R code:

1. $U[0,1]$ vs $U[0,1]$

```
>install.packages("AR")
>library(AR)
> simulation = AR.Sim( n = 10,
  f_X = function(y){dunif(y,min=0,max=1)},
  Y.dist = "unif",
  Y.dist.par = c(0,1),
  Rej.Num = TRUE,
  Rej.Rate = TRUE,
  Acc.Rate = FALSE)
>simulation
```

2. Uniform (0,1) VS Standard Normal (0,1)

```
simulation = AR.Sim( n = 10,
  f_X = function(y){dunif(y,min=0,max=1)},
  Y.dist = "norm",
  Y.dist.par = c(0,1),
  Rej.Num = TRUE,
  Rej.Rate = TRUE,
  Acc.Rate = FALSE)
>simulation
```

3. Uniform (0,1) VS Normal (mean,var)

```
simulation = AR.Sim( n = 10,
  f_X = function(y){dunif(y,min=0,max=1)},
  Y.dist = "norm",
  Y.dist.par = c(mean,s.d),
  Rej.Num = TRUE,
  Rej.Rate = TRUE,
  Acc.Rate = FALSE)
>simulation
```

1. Example: Write R Program to simulate the data for $n=8$ using Acceptance-Rejection Algorithm for

Uniform distribution in the interval (0,1) and Normal Distribution with parameters mean 10 and variance 36 and Comment on your result.

R Code with Output:

```
install.packages("AR")
```

```
library(AR)
```

```
simulation = AR.Sim( n = 8,
```

```
    f_X = function(y){dunif(y,min=0,max=1)},
```

```
    Y.dist = "norm",
```

```
    Y.dist.par = c(10,6),
```

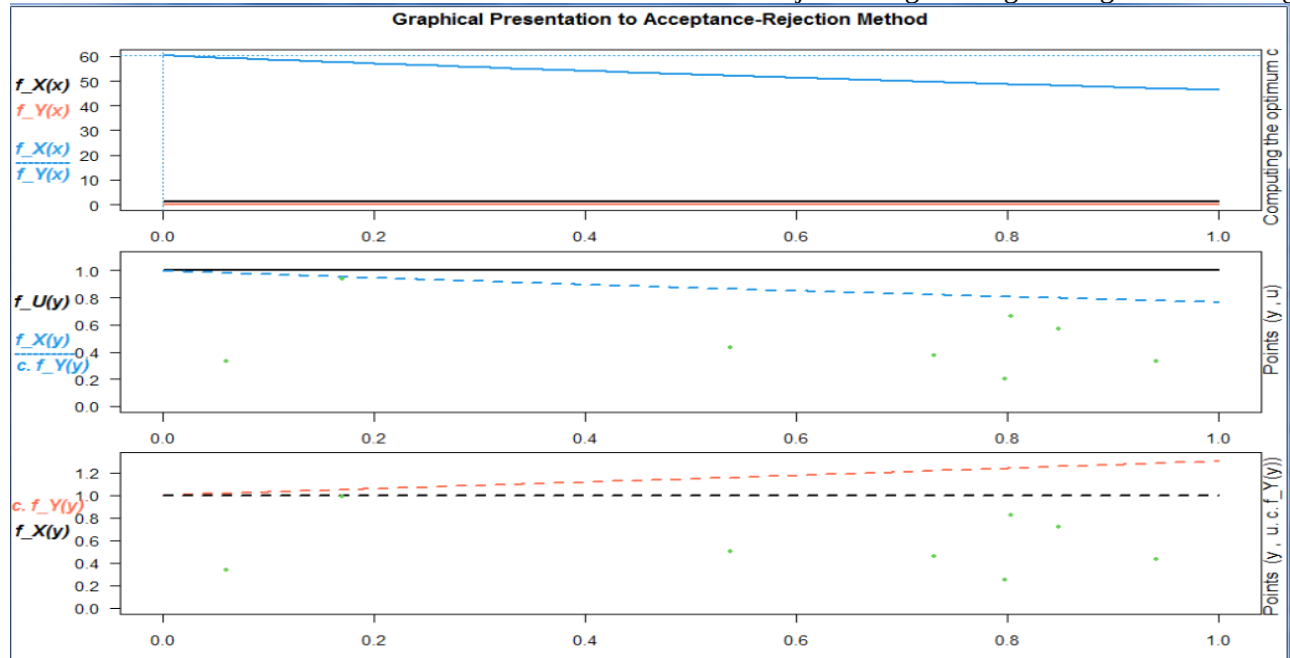
```
    Rej.Num = TRUE,
```

```
    Rej.Rate = TRUE,
```

```
    Acc.Rate = FALSE)
```

```
simulation
```

```
> simulation = AR.Sim( n = 8,
+                      f_X = function(y){dunif(y,min=0,max=1)},
+                      Y.dist = "norm",
+                      Y.dist.par = c(10,6),
+                      Rej.Num = TRUE,
+                      Rej.Rate = TRUE,
+                      Acc.Rate = FALSE)
Optimal c = 60.314
The numbers of Rejections = 201
Ratio of Rejections = 0.962
> simulation
[1] 0.73086822 0.80376564 0.79708956 0.94135794 0.84846397 0.53779598 0.06042864 0.17044630
```



Inference: In this case, the rejection ration is 0.96 is significantly higher when sampling from Uniform distribution within the interval 0,1.

2. Example: Write R Program to generate 8 observations using Acceptance-Rejection Algorithm for Uniform distribution in the interval (0,1) and Standard Normal Distribution and Comment on your result.

R Code with Output:

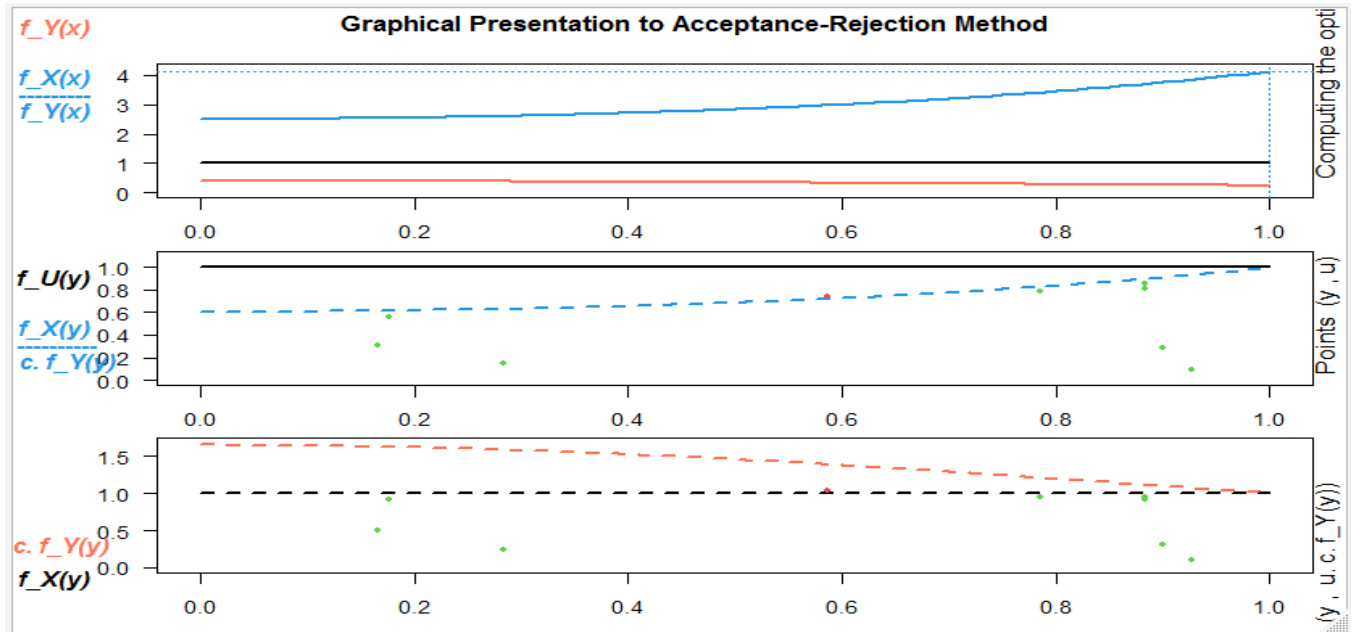
```
simulation = AR.Sim( n = 8,
  f_X = function(y){dunif(y,min=0,max=1)},
  Y.dist = "norm",
  Y.dist.par = c(0,1),
  Rej.Num = TRUE,
  Rej.Rate = TRUE,
  Acc.Rate = FALSE)

simulation
```

```

> simulation = AR.Sim( n = 8,
+                     f_X = function(y){dunif(y,min=0,max=1)},
+                     Y.dist = "norm",
+                     Y.dist.par = c(0,1),
+                     Rej.Num = TRUE,
+                     Rej.Rate = TRUE,
+                     Acc.Rate = FALSE)
Optimal c = 4.132
The numbers of Rejections = 10
Ratio of Rejections = 0.556
> simulation
[1] 0.7860279 0.8839270 0.1767668 0.9269260 0.1661684 0.2832707 0.8832648
[8] 0.9009023

```



Inference: In this case, the rejection ration is 0.56 is not much significantly higher when sampling from Uniform distribution within the interval 0,1.

3. Example: Write R Program to generate 5 observations using Acceptance-Rejection Algorithm for Uniform distribution in the interval (0,1) and Uniform distribution in the interval (0,1) and Comment on your result.

R Code with Output:

```

simulation = AR.Sim( n = 5,
                     f_X = function(y){dunif(y,min=0,max=1)},
                     Y.dist = "unif",
                     Y.dist.par = c(0,1),
                     Rej.Num = TRUE,
                     Rej.Rate = TRUE,
                     Acc.Rate = FALSE)

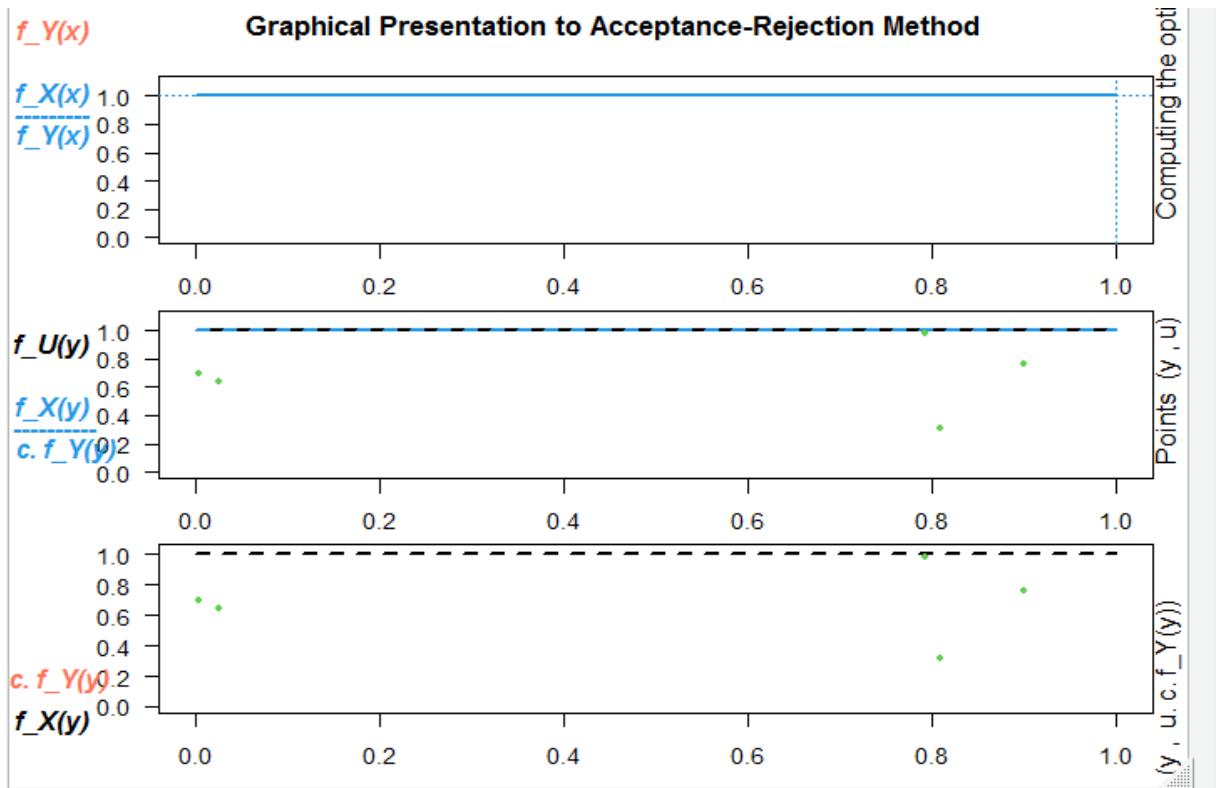
```

simulation

```

> simulation = AR.Sim( n = 5,
+                     f_X = function(y){dunif(y,min=0,max=1)},
+                     Y.dist = "unif",
+                     Y.dist.par = c(0,1),
+                     Rej.Num = TRUE,
+                     Rej.Rate = TRUE,
+                     Acc.Rate = FALSE)
Optimal c = 1
The numbers of Rejections = 0
Ratio of Rejections = 0
> simulation
[1] 0.809260487 0.002256192 0.900072686 0.025223927 0.792977998

```



Inference: In this case, the rejection ration is 0. i.e generated random numbers are from same Uniform distribution. Hence all random numbers fall within the specified distribution.

4. Maximum likelihood and Method of moment's estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R

4.1 Maximum likelihood Estimation:

R Code:

```
> x=c(data vector)
> n=length(x)           #No. of observations
> phat=mean(x)/n         #MLE for Binomial distribution
where n represents no.of independent trails in Binomial distribution
> lambdahat=mean(x)      #MLE for Poisson distribution
> muhat=mean(x)          #MLE of Mean for Normal distribution
> sigmahat=((n - 1)/n)*var(x) #MLE of variance for Normal distribution
```

Note: If the sample data is not given then generate the sample using random numbers of a particular distribution and then estimate the corresponding parameters.

R Code to generate the random numbers from Binomial, Poisson and Normal distribution

rbinom(no of random numbers to be generate, n, p)

rpois(n,lambda)

rnorm(n, mean, sd)

1. Example: Sample (9,2,5,3,1,5,2,4,2,1,3,5,6,1,1) is drawn from Binomial distribution. Use R program

to find the maximum likelihood estimate of the parameter p and estimate the mean and variance.

R Code with Output:

```
> x=c(9,2,5,3,1,5,2,4,2,1,3,5,6,1,1)
> phat=mean(x)/length(x)
> phat
[1] 0.2222222
> meanhat=length(x)*phat
> meanhat
[1] 3.333333
> varhat=length(x)*phat*(1-phat)
> varhat
[1] 2.592593
```

2. Example: Let X follows Poisson distribution with parameter 12. Use R program to generate 10 samples from Poisson distribution and hence find the maximum likelihood estimate of the parameter

and estimate the mean and variance.

R Code with Output:

```
> x=rpois(10,12)
> lambdahat=mean(x)
> lambdahat
[1] 9.3
> meanhat=lambdahat
> meanhat
[1] 9.3
> varhat=lambdahat
> varhat
[1] 9.3
```

3. Example: Sample (15, 9, 3, 7, 15, 8, 10, 3, 12, 5, 15, 8, 10, 3, 15, 8, 10, 3, 15, 8, 10, 3, 15, 7, 14, 10, 10, 14, 10, 9, 3, 3, 10, 15, 8, 10, 3, 15, 8, 10, 3, 12, 5, 15, 8, 10, 3, 15, 8, 10, 3) is drawn from Normal distribution. Use R program to estimate the mean and variance using maximum likelihood estimation.

R Code with Output:

```
> x=c(15, 9, 3, 7, 15, 8, 10, 3, 12, 5, 15, 8, 10, 3, 15, 8, 10, 3, 15, 8, 10, 3, 15, 7, 14, 10, 10, 14, 10, 9, 3, 3, 10, 15, 8, 10, 3, 15, 8, 10, 3, 12, 5, 15, 8, 10, 3, 15, 8, 10, 3)
> muhat=mean(x)
> muhat
[1] 9.039216
> n=length(x)
> sigmahat=((n - 1)/n)*var(x)
> sigmahat*sigmahat
[1] 300.3911
```

4.2 Method of Moments estimation

R Code

```
> x=c(data vector)
```

```
>n=length(x)           #No. of observations
```

```
>phat=mean(x)/n        #MLE for Binomial distribution
```

where **n** represents no.of independent trails in Binomial distribution

```
>lambdahat=mean(x)      #MLE for Poisson distribution
```

```
>muhat=mean(x)          #MLE of Mean for Normal distribution
```

```
>sigmahat=((n - 1)/n)*var(x) #MLE of variance for Normal distribution
```

1. Example: Sample (9, 3, 7, 12, 5, 15, 15, 8, 10, 3, 15, 7, 14, 10, 10, 14, 10, 9, 3, 3, 10) is drawn from

Poisson distribution. Use R program to estimate the parameter, mean and variance of Poisson using Method of Moments.

R Code with Output:

```
> x=c(9, 3, 7, 12, 5, 15, 15, 8, 10, 3, 15, 7, 14, 10, 10, 14, 10, 9, 3, 3, 10)
> lambdahat=mean(x)
> lambdahat
[1] 9.142857
> meanhat=lambdahat
> meanhat
[1] 9.142857
> varhat=lambdahat
> varhat
[1] 9.142857
```

2. Example: Sample (12, 5, 10, 11, 14, 9, 3, 6, 8, 12, 14, 10, 9, 3, 2, 4, 6, 3, 10) is drawn from Binomial distribution. Use R program to estimate the parameter, mean and variance of Binomial using Method of Moments.

R Code with Output:

```
> x=c(12, 5, 10, 11, 14, 9, 3, 6, 8, 12, 14, 10, 9, 3, 2, 4, 6, 3, 10)
> phat=mean(x)/length(x)
> phat
[1] 0.4182825
> meanhat=length(x)*phat
> meanhat
[1] 7.947368
> varhat=length(x)*phat*(1-phat)
> varhat
[1] 4.623123
```

3. Example: Let X follows Binomial Distribution with parameters $n=20$ and $p=0.36$. Use R program to generate 8 random samples from Binomial distribution and hence estimate the parameter p , mean and variance of Binomial using Method of Moments.

[Hint: To estimate the parameter $\text{phat}=\text{mean}(\text{datavector})/\text{no.of independent trails}$]

R Code with Output:

```
> x=rbinom(8,20,0.36)
> x
[1] 9 9 7 9 3 10 11 9
> phat=mean(x)/20
> phat
[1] 0.41875
> meanhat=20*phat
> meanhat
[1] 8.375
> varhat=20*phat*(1-phat)
> varhat
[1] 4.867969
```

4.3 Testing of Hypothesis

a. Compute one sample proportion of z-test

R Code:

prop.test(x, n, p, alternative = "two.sided")

prop.test(x, n, p, alternative = "less")

prop.test(x, n,p, alternative = "greater")

x: the number of successes

n: the total number of trials

p: the probability to test against.

1. Example: A manufacture of spark plugs claims that 3% of items supplied by him defective. A random sample of 500 plugs is found to have 20 defectives items. test the claim of manufacture.

R Code with Output:

prop.test(20,500,0.03,alternative="two.sided")

From the output we observe that:

Null Hypothesis (H_0): $p = 0.03$

Alternative Hypothesis (H_1): $p \neq 0.03$

And the p value is $0.2381 > 0.05$ level of significance

Hence We Accept Null hypothesis

95 % Confidence interval is (0.0252, 0.0621)

Phat = 0.04

2. Example: We have a population of mice containing half male and have female ($p = 0.5 = 50\%$). Some of these mice ($n = 160$) have developed a spontaneous cancer, including 95 male and 65 female. Test whether the cancer affects more male than female?

R Code with Output:

```
prop.test(95,160,0.5,alternative="greater")
```

From the output we observe that:

Null Hypothesis (H_0): $p = 0.5$

Alternative Hypothesis (H_1): $p > 0.5$

And the p value is $0.01093 < 0.05$ level of significance (los)

Hence We Reject Null hypothesis

95 % Confidence interval is (0.525, 1.0)

Phat = 0.593

b. Compute two-proportions of z-test

R Code:

```
prop.test(x=c(.....), n=c(.....), p = NULL, alternative = ".....")
```

1. Example: In a large city A, consists of 900 school children, 20 had defective eyesight. In other large city B of a random sample 600 children, 15 had the same defect. Is this difference between the two proportions significant?

R Code with Output:

```
prop.test(x=c(900,600),n=c(20,15),p=NULL,alternative="two.sided")
```

From the output we observe that:

Null Hypothesis (H_0): $p_1 = p_2$

Alternative Hypothesis (H_1): $p_1 \neq p_2$

And the p value is $0.8614 > 0.05$ level of significance

Hence We Accept Null hypothesis

95 % Confidence interval is (-0.019, 0.014)

P1hat = 0.0222 P2hat=0.025

2. Example: In a company 500 people are having lung cancer and 500 people are healthy individuals. The number of smokers in each group is 490 and 400. Test whether the proportions of smokers are the same in the two groups of individuals?

R Code with Output:

```
prop.test(x=c(490,400),n=c(500,500),p=NULL,alternative="two.sided")
```

From the output we observe that:

Null Hypothesis (H_0): $p_1=p_2$

Alternative Hypothesis (H_1): $p_1 \neq p_2$

And the p value is $2.2e-16 < 0.05$ level of significance

Hence We Reject Null hypothesis

95 % Confidence interval is (0.1408, 0.2191)

$P_{1hat} = 0.98$ $P_{2hat} = 0.80$

C. One sample Z test in R

R Code:

```
install.packages("BSDA") #BSDA: Basic Statistics and Data Analysis
```

```
library(BSDA)
```

```
z.test(datavector, mu=population mean, sigma.x=....., alternative= ".....")
```

1. Example: The price of a popular tennis racket at a national chain store is \$179. Portia bought ten of the same racket at an online auction site for the following prices:

155,179,175,170,175,173,172,171,170,169,168,164,165,162,163,172,176,174,168,165,170,172,168,

162,175,161,160,158,156,171,170. If the auction prices of rackets are normally distributed, Use R code to determine whether there is sufficient evidence in the sample, at the 5% level of significance, to

conclude that the average price of the racket is less than \$179 with standard deviation 6.1 if purchased at an online auction.

R Code with Output:

```
install.packages("BSDA") #BSDA: Basic Statistics and Data Analysis
```

```
library(BSDA)
```

```
x=c(155,179,175,170,175,173,172,171,170,169,168,164,165,162,163,172,176,174,168,165,170,172,168,162,175,161,160,158,156,171,170)
```

```
z.test(x,mu=179,sigma.x=6.1,alternative="less")
```

From the output we observe that:

Null Hypothesis (H_0): $\mu=179$

Alternative Hypothesis (H_1): $\mu < 179$

And $z = -10.011$; the p value is $2.2e-16 < 0.05$ level of significance

Hence We Reject Null hypothesis

95 % Confidence interval is (NA, 169.834)

Meanhat=168.032

d.Two Sample Z-Test in R

R Code:

z.test(datavector1,datavector2, alternative='.....', mu=0, sigma.x=....., sigma.y=.....)

1. Example: Suppose the IQ levels among individuals in two different cities are known to be normally distributed each with population standard deviations of 15 and 12 respectively. A scientist wants to test whether the mean IQ level between individuals in city A and city B are different, so she selects a simple random sample of 32 from city A and 36 samples from city B and records their IQ levels.

[Hint: First generate random samples using rnorm]

R Code with Output:

cityA=rnorm(32,0,15)

cityB=rnorm(36,0,12)

z.test(cityA,cityB,mu=0,sigma.x=15,sigma.y=12,alternative="two.sided")

From the output we observe that:

Null Hypothesis (H_0): $\mu_1 = \mu_2$

Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$

And $z = -1.7913$; the p value is $0.073 > 0.05$ level of significance

Hence We Accept Null hypothesis

95 % Confidence interval is (-12.459, 0.560)

Meanhat1= -0.738 Meanhat2= 5.211

Viva Questions:

1. Define Null Hypothesis and Alternative Hypothesis.

Null Hypothesis (H_0): A null hypothesis is the hypothesis which asserts that there is no significant difference between the sample and the population parameter.

Alternative Hypothesis (H_1): Any hypothesis which contradicts the Null hypothesis is called Alternative hypothesis.

2. Write the test statistic to test the difference of two proportions.

$$z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim N(0,1)$$

3. Write the test-statistic to test for single mean in large samples.

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$$

5. Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R

5.1 t test

i) t-test for single mean

R Code:

t.test(data vector, mu = population mean, alternative= "two.sided")

t.test(data vector, mu = population mean, alternative= "less")

t.test(data vector, mu = population mean, alternative= "greater")

1. **Example:** A random sample of 8 envelopes is taken from the letter box of a post office and their weights in grams are found to be 12.1, 11.9, 12.4, 12.3, 11.5, 11.6, 12.1 and 12.4. Does this sample indicate at 1% level that the average weight of envelopes received at their post office is 12.35 grams? Also construct 95% confidence interval for population mean.

R Code with Output:

x=c(12.1,11.9,12.4,12.3,11.5,11.6,12.1,12.4)

t.test(x,mu=12.35,alternative="two.sided")

From the output we observe that:

Null Hypothesis (H_0): $\mu_1 = 12.35$

Alternative Hypothesis (H_1): $\mu_1 \neq 12.35$

And $t = -2.553$; the p value is $0.037 < 0.05$ level of significance

Hence We Reject Null hypothesis

95 % Confidence interval is (11.748, 12.326)

Meanhat= 12.037

2. **Example:** Test the hypothesis that the average content of containers of a particular lubricant is 10 liters if the content of a random sample of 10 containers are:

10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, 9.8

Use 0.01 l.o.s and assume that the distribution of contents is normal. Also construct 95% confidence interval for population mean.

_R Code with Output:

x=c(10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, 9.8)

t.test(x,mu=10,alternative="two.sided")

From the output we observe that:

Null Hypothesis (H_0): $\mu_1 = 10$

Alternative Hypothesis (H_1): $\mu_1 \neq 10$

And $t = 0.771$; the p value is $0.46 > 0.05$ level of significance

Hence We Accept Null hypothesis

95 % Confidence interval is (9.884, 10.235)

Meanhat= 10.06

ii) Two sample t test

R Code:

t.test(data vector 1, data vector 2, mu = 0, alternative= “.....”)

1. **Example:** Two Horses A and B were tested according to the time (seconds) in a particular track with the following results.

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	-

Test whether the two horses have same running capacity? Also construct 95% confidence interval for population mean.

R Code with Output:

HorseA=c(28,30,32,33,33,29,34)

HorseB=c(29,30,30,24,27,29)

t.test(HorseA,HorseB,mu=0,alternative="two.sided")

From the output we observe that:

Null Hypothesis (H_0): $\mu_1 = \mu_2$

Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$

And $t = 2.4335$; the p value is $0.033 < 0.05$ level of significance

Hence, We Reject Null hypothesis

95 % Confidence interval is (0.286, 5.951)

Meanhatx= 31.285 Meanhaty= 28.166

iii) Paired t test**R Code:**

```
t.test(after, before, paired = TRUE, alternative = ".....")
```

1. Example: Memory capacity of 10 students was tested before and after training. Test whether the training was effective or not from the following scores. Also construct 95% confidence interval for population mean.

Before training	12	14	11	8	7	10	3	0	5	6
After training	15	16	10	7	5	12	10	2	3	8

R Code with Output:

```
before=c(12,14,11,8,7,10,3,0,5,6)
```

```
after=c(15,16,10,7,5,12,10,2,3,8)
```

```
t.test(before,after,paired=TRUE,alternative="two.sided")
```

From the output we observe that:

Null Hypothesis (H_0): $\mu_1 = \mu_2$

Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$

And $t = -1.3646$; the p value is $0.2055 > 0.05$ level of significance

Hence, We Accept Null hypothesis

95 % Confidence interval is $(-3.189, 0.789)$

Meandifference = -1.2

5.2 Chi-square test**R Code:**

```
x=matrix(c(data),ncol=.....,nrow=.....,byrow=TRUE)
```

or

```
x=data.frame(col1 = c (.....), col2 = c(.....), .....)
```

```
x
```

```
chisq.test(x)
```

1. Example: The following data gives the classification of 100 workers according to gender and nature

of work. Test whether the nature of work is independent of the gender of the worker.

	Stable	Unstable
Males	40	20
Females	10	30

R Code with Output

```
x=matrix(c(40,20,10,30),ncol=2,nrow=2,byrow=TRUE)
x
chisq.test(x)
```

From the output we observe that:

Null Hypothesis (H_0): Two attributes are independent

Alternative Hypothesis (H_1): Two attributes are not independent

And chi-square = 15.042; the p value is $0.0001 < 0.05$ level of significance

Hence, We Reject Null hypothesis

2. **Example:** 1000 students at college level were graded according to their I.Q and the economic conditions of their home. Use χ^2 test to find whether there is any association between condition at home and I.Q

	I.Q		
Economic conditions	High	Low	Total
Rich	460	140	600
Poor	240	160	400
Total	700	300	1000

R Code with Output

```
x=matrix(c(460,140,240,160),ncol=2,nrow=2,byrow=TRUE)
x
chisq.test(x)
```

From the output we observe that:

Null Hypothesis (H_0): Two attributes are independent

Alternative Hypothesis (H_1): Two attributes are not independent

And chi-square = 30.957; the p value is $0.0000 < 0.05$ level of significance

Hence, We Reject Null hypothesis

5.3 F-test.

R Code:

```
var.test(sample1,sample2,alternative="....")
```

Note: While calculating F ratio, F value should be greater than 1. If not interchange the samples.

1. **Example:** Two Horses A and B were tested according to the time (seconds) in a particular track with the following results.

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	-

Test at 5% level whether the populations have the same variance.

R Code with Output:

```
x=c(28,30,32,33,33,29,34)
y=c(29,30,30,24,27,29)
var.test(y,x,alternative="two.sided")
```

From the output we observe that:

Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$

Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$

And $F = 1.0245$; the p value is $0.9573 > 0.05$ level of significance

Hence, We Accept Null hypothesis

95 % Confidence interval is (0.1711, 7.148)

2. **Example:** The following are the values in thousands of an inch obtained by 2 engineers in 8 successive measurements with the same micrometer:

Engineer A: 503,505,497,505,419,493,510,501

Engineer B: 502,497,492,490,495,497,596,498

Is one engineer more consistent than the other?

R Code with Output:

```
x=c(503,505,497,505,419,493,510,501)
y=c(502,497,492,490,495,497,596,498)
var.test(y,x,alternative="two.sided")
```

From the output we observe that:

Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$

Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$

And $F = 1.4265$; the p value is $0.651 > 0.05$ level of significance

Hence, We Accept Null hypothesis

95 % Confidence interval is (0.285, 7.125)

6. Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi collinearity in R

6.1 Fitting of Multiple Linear Regression

R Code:

```
MLR<- lm(Dependent variable ~ sum of
independent variables)
```

➤ Testing MLR

R Code:

```
summary(MLR)
```

6.2. Outlier analysis

R Code:

```
plot(MLR)
```

6.3. Residual analysis

R Code

```
Residuals=resid(MLR)
Predicted=predict(MLR)
Residuals
Predicted
plot(Residuals,Predicted)
```

6.4. Test for normality

R Code:

```
plot(MLR,2)
```

2 indicates second plot i.e Normal Q-Q plot

```
shapiro.test(Residuals)
```

to test the Residuals follows normality or not

6.5. Multi Collinearity

R Code:

```
install.packages("caTools")
```

For Linear regression

```
install.packages('car')
```

To check multicollinearity

```
install.packages("quantmod")
```

```
install.packages("MASS")
```

```
install.packages("corrplot")
```

plot correlation plot

```
library(caTools)
```

```
library(car)
```

```
library(quantmod)
```

```
library(MASS)
```

```
library(corrplot)
```

```
vif(MLR) #Variance Inflation Factor (vif)
```

specifies whether Multicollinearity present in the variables or not

if vif value is >10 then there exist Multicollinearity

1. Example:

The diameter of a tree at its breast height is influenced by many factors. The below table relate the diameter of a particular type of eucalyptus tree to its age, average rainfall at its site, site's elevation, and the wood's mean specific gravity.

S.No.	Age (Years)	Elevation (1000 ft)	Rainfall (inches)	Specific Gravity	Diameter at Breast Height (inches)
1	44	1.3	250	0.63	18.1
2	33	2.2	115	0.59	19.6
3	33	2.2	75	0.56	16.6
4	32	2.6	85	0.55	16.4
5	34	2.0	100	0.54	16.9
6	31	1.8	75	0.59	17.0
7	33	2.2	85	0.56	20.0
8	30	3.6	75	0.46	16.6
9	34	1.6	225	0.63	16.2
10	34	1.5	250	0.60	18.5
11	33	2.2	255	0.63	18.7
12	36	1.7	175	0.58	19.4
13	33	2.2	75	0.55	17.6
14	34	1.3	85	0.57	18.3
15	37	2.6	90	0.62	18.8

- Fit a Multiple linear regression model and Test the Model
- Identify the Outliers in the data
- Check for Multicollinearity for the variables
- Compute the Residual Analysis
- Test for Normality

a) R Code with Output:

Diameter=c(18.1,19.6,16.6,16.4,16.9,17,20,16.6,16.2,18.5,18.7,19.4,17.6,18.3,18.8)

Age=c(44,33,33,32,34,31,33,30,34,34,33,36,33,34,37)

Elevation=c(1.3,2.2,2.2,2.6,2,1.8,2.2,3.6,1.6,1.5,2.2,1.7,2.2,1.3,2.6)

Rainfall=c(250,115,75,85,100,75,85,75,225,250,255,175,75,85,90)

Gravity=c(0.63,.59,.56,.55,.54,.59,.56,.46,.63,.60,.63,.58,.55,.57,.62)

MLR=lm(Diameter~Age+Elevation+Rainfall+Gravity)

summary(MLR)

```
call:
lm(formula = Diameter ~ Age + Elevation + Rainfall + Gravity)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.9920	-0.8148	-0.1088	0.5898	2.2203

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.855179	8.755170	1.240	0.243
Age	0.063384	0.145366	0.436	0.672
Elevation	0.074448	0.856266	0.087	0.932
Rainfall	-0.001469	0.006874	-0.214	0.835
Gravity	8.560628	13.012268	0.658	0.525

Residual standard error: 1.391 on 10 degrees of freedom

Multiple R-squared: 0.12, Adjusted R-squared: -0.232

F-statistic: 0.3409 on 4 and 10 DF, p-value: 0.8444

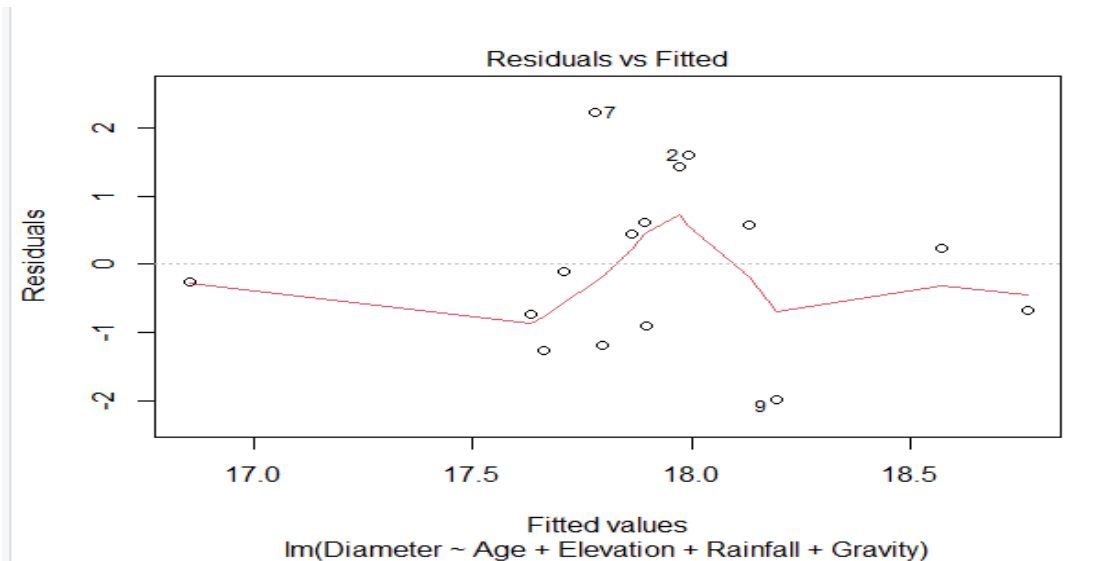
Inference:

$$\text{Diameter} = 10.855 + 0.063 * \text{Age} + 0.074 * \widehat{\text{Elevation}} - 0.001 * \text{Rainfall} + 8.560 * \text{Gravity}$$

If p-value = 0.844 > 0.05 level of significance then we Accept H_0 .

b) R Code with Output:

plot(MLR)



Inference: From the above graph 2,7,9 are outliers

c) R Code with Output:

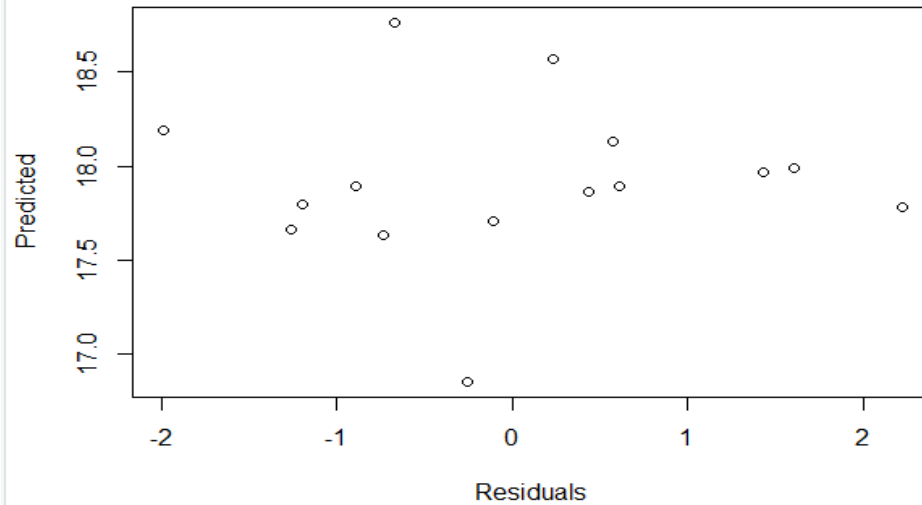
```
install.packages("caTools") # For Linear regression
install.packages('car') # To check multicollinearity
install.packages("quantmod")
install.packages("MASS")
install.packages("corrplot") # plot correlation plot
library(caTools)
library(car)
library(quantmod)
library(MASS)
library(corrplot)
vif(MLR)
> vif(MLR)
      Age Elevation Rainfall Gravity
1.605384 1.870223 1.859050 2.476998
```

Inference:

No factor is having >10. Hence there does not exist multi collinearity.

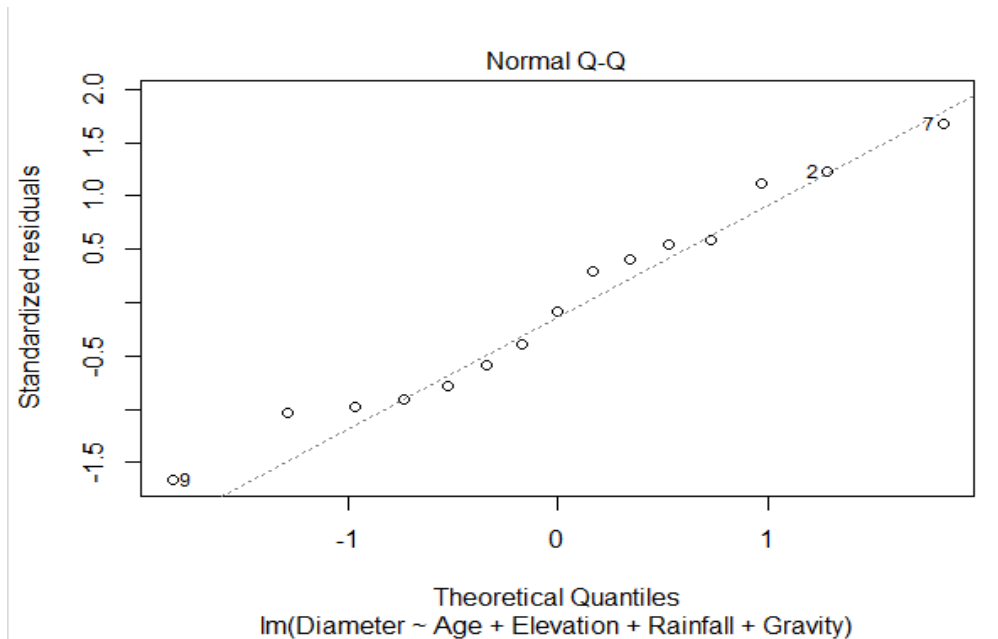
d) R Code with Output:

```
Residuals=resid(MLR)
Predicted=predict(MLR)
Residuals
Predicted
plot(Residuals,Predicted)
> Residuals=resid(MLR)
> Predicted=predict(MLR)
> Residuals #displays residuals= Actual - predicted values
      1      2      3      4      5      6      7
-0.6668225 1.6075154 -1.1944245 -1.2605235 -0.7349824 -0.8946956 2.2202652
      8      9     10     11     12     13     14
-0.2524362 -1.9920384 0.6089495 0.5707462 1.4283311 -0.1088183 0.4382778
     15
0.2306560
> Predicted #displays predicted values
      1      2      3      4      5      6      7      8      9
18.76682 17.99248 17.79442 17.66052 17.63498 17.89470 17.77973 16.85244 18.19204
     10     11     12     13     14     15
17.89105 18.12925 17.97167 17.70882 17.86172 18.56934
> plot(Residuals,Predicted)
```



e) R Code with Output:

```
plot(MLR,2)
```



```
shapiro.test(Residuals)
```

```
> shapiro.test(Residuals)
```

```
shapiro-wilk normality test
```

```
data: Residuals
```

```
w = 0.97799, p-value = 0.9539
```

Inference: If $p\text{-value} = 0.9539 > 0.05$

Accept H_0 .

Viva Questions

1. Define Linear Regression.

It is the mathematical average relationship between dependent and independent variables.

2. Define Residual.

Difference between actual response variable and predicted response variable.

3. Define Multicollinearity and when Multicollinearity exist in linear regression.

Multicollinearity occurs when two or more independent variables in a data frame have a high correlation with one another in a regression model.

Multicollinearity exists whenever an independent variable is highly correlated with one or more of the other independent variables in a multiple regression equation.

7.Logistic Regression in R**R Code:**

```
Logistic_Reg = glm(Dependent variable ~ sum of Independent variables, family="binomial")  
summary(Logistic_Reg)
```

1. Example:

Consider data published on $n = 27$ leukemia patients. The data has a response variable of whether leukemia remission occurred (REMISS), which is given by a 1. The predictor variables are cellularity of the marrow clot section (CELL), smear differential percentage of blasts (SMEAR), percentage of absolute marrow leukemia cell infiltrate (INFIL), percentage labeling index of the bone marrow leukemia cells (LI), absolute number of blasts in the peripheral blood (BLAST), and the highest temperature prior to start of treatment (TEMP).

REMISS	CELL	SMEAR	INFIL	LI	BLAST	TEMP
1	0.8	0.83	0.66	1.9	1.1	1
1	0.9	0.36	0.32	1.4	0.74	0.99
0	0.8	0.88	0.7	0.8	0.18	0.98
0	1	0.87	0.87	0.7	1.05	0.99
1	0.9	0.75	0.68	1.3	0.52	0.98
0	1	0.65	0.65	0.6	0.52	0.98
1	0.95	0.97	0.92	1	1.23	0.99
0	0.95	0.87	0.83	1.9	1.35	1.02
0	1	0.45	0.45	0.8	0.32	1
0	0.95	0.36	0.34	0.5	0	1.04
0	0.85	0.39	0.33	0.7	0.28	0.99
0	0.7	0.76	0.53	1.2	0.15	0.98
0	0.8	0.46	0.37	0.4	0.38	1.01
0	0.2	0.39	0.08	0.8	0.11	0.99
0	1	0.9	0.9	1.1	1.04	0.99
1	1	0.84	0.84	1.9	2.06	1.02
0	0.65	0.42	0.27	0.5	0.11	1.01
0	1	0.75	0.75	1	1.32	1
0	0.5	0.44	0.22	0.6	0.11	0.99
1	1	0.63	0.63	1.1	1.07	0.99
0	1	0.33	0.33	0.4	0.18	1.01
0	0.9	0.93	0.84	0.6	1.59	1.02
1	1	0.58	0.58	1	0.53	1
0	0.95	0.32	0.3	1.6	0.89	0.99
1	1	0.6	0.6	1.7	0.96	0.99
1	1	0.69	0.69	0.9	0.4	0.99
0	1	0.73	0.73	0.7	0.4	0.99

R Code with Output:

```
Remiss=c(1,1,0,0,1,0,1,0,0,0,0,0,0,0,1,0,0,0,1,0,0,1,0,1,1,0)
cell=c(.8,.9,.8,1,.9,1,.95,.95,1,.95,.85,.7,.8,.2,1,1,.65,1,.5,1,1,.9,1,.95,1,1,1)
smear=c(.83,.36,.88,.87,.75,.65,.97,.87,.45,.36,.39,.76,.46,.39,.9,.84,.42,.75,.44,.63,.33,.93,.58,.32,.6,.69,.73)
infil=c(.66,.32,.7,.89,.68,.65,.92,.83,.45,.34,.33,.53,.37,.08,.9,.84,.27,.75,.22,.63,.33,.84,.58,.3,.6,.69,.73)
l1=c(1.9,1.4,0.8,.7,1.3,.6,1.1,1.9,.8,.5,.7,1.2,.4,.8,1.1,1.9,.5,1,.6,1.1,.4,.6,1.1,1.6,1.7,0.9,0.7)
blast=c(1.1,.74,.18,1.05,0.52,0.52,1.23,1.35,0.32,0,0.28,0.15,0.38,0.11,1.04,2.06,0.11,1.32,0.11,1.07,0.18,1.59,0.53,0.89,0.96,0.4,0.4)
temp=c(1,.99,.98,.99,.98,.99,.99,1.02,1,1.04,0.99,.98,1.01,.99,.99,1.02,1.01,1,.99,.99,1.01,1.02,1,.99,.99,.99,.99)
logistic_Model=glm(Remiss~cell+smear+infil+l1+blast+temp,family="binomial")
logistic_Model
```

```
Call: glm(formula = Remiss ~ cell + smear + infil + l1 + blast + temp,
  family = "binomial")

Coefficients:
(Intercept)      cell      smear      infil      l1      blast
  74.704      43.501      38.723     -39.902      4.354      0.359
    temp
 -122.924

Degrees of Freedom: 26 Total (i.e. Null);  20 Residual
Null Deviance:      34.37
Residual Deviance: 20.76      AIC: 34.76
```

Inference:

$$\widehat{Remiss} = 74.704 + 43.501 * cell + 38.723 * smear - 39.902 * infil + 4.354 * l1 + 0.359 * blast - 122.924 * temp$$

summary(logistic_Model)

```
Call:
glm(formula = Remiss ~ cell + smear + infil + l1 + blast + temp,
     family = "binomial")
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.08053	-0.61978	-0.01379	0.75830	1.55672

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	74.704	74.685	1.000	0.317
cell	43.501	48.801	0.891	0.373
smear	38.723	56.436	0.686	0.493
infil	-39.902	59.808	-0.667	0.505
l1	4.354	2.725	1.598	0.110
blast	0.359	2.319	0.155	0.877
temp	-122.924	82.645	-1.487	0.137

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 34.372 on 26 degrees of freedom
 Residual deviance: 20.761 on 20 degrees of freedom
 AIC: 34.761

Number of Fisher Scoring iterations: 8

Inference:

Among all p-values only l1 i.e Index of the bone marrow leukemia cells appears to be closed to a significant predictor of remission occurring and the next factor is closest to response is temp i.e highest temperature prior to start of treatment.

Viva Questions:

1. What kind of Machine Learning is Logistic Regression?

Supervised Machine Learning

2. Define Binary Response Variable?

A binary response variable is a categorical variable that can only take one of two values

3. State any two differences of Linear Regression and Logistic Regression?

Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic Regression is used to predict the categorical dependent variable using a given set of independent variables.
Linear Regression is used for solving Regression problem.	Logistic regression is used for solving Classification problems.

8. Classification with SVM in R

R Code:

```
install.packages("e1071")
library(e1071)
x = matrix(rnorm(32), 16, 2)      #Generating the data
y = rep(c(-1, 1), c(8, 8))        #Assigning +1 and -1 to the above data
x[y == 1,] = x[y == 1,] + 1      #bias the x variable when y =1
dat = data.frame(x, y = as.factor(y)) #representing the above data in table form using dataframe
svm = svm(y ~ ., data = dat, kernel = "linear", cost=10)
summary(svm)
```

1. Example: Write R code to find the number of support vectors for classifying the data. Data as follows: 60 observations, in two columns and also get the SVM classification plot.

R Code with Output:

```
install.packages("e1071")
library(e1071)
x = matrix(rnorm(60), 30, 2)
y = rep(c(-1, 1), c(15, 15))
x[y == 1,] = x[y == 1,] + 1
data = data.frame(x, y = as.factor(y))
svm = svm(y ~ ., data = data, kernel = "linear", cost=10)
summary(svm)
```

```
Call:
svm(formula = y ~ ., data = data, kernel = "linear", cost = 10)
```

```
Parameters:
```

```
  SVM-Type:  c-classification
SVM-Kernel:  linear
      cost:  10
```

```
Number of Support Vectors:  13
```

```
( 6 7 )
```

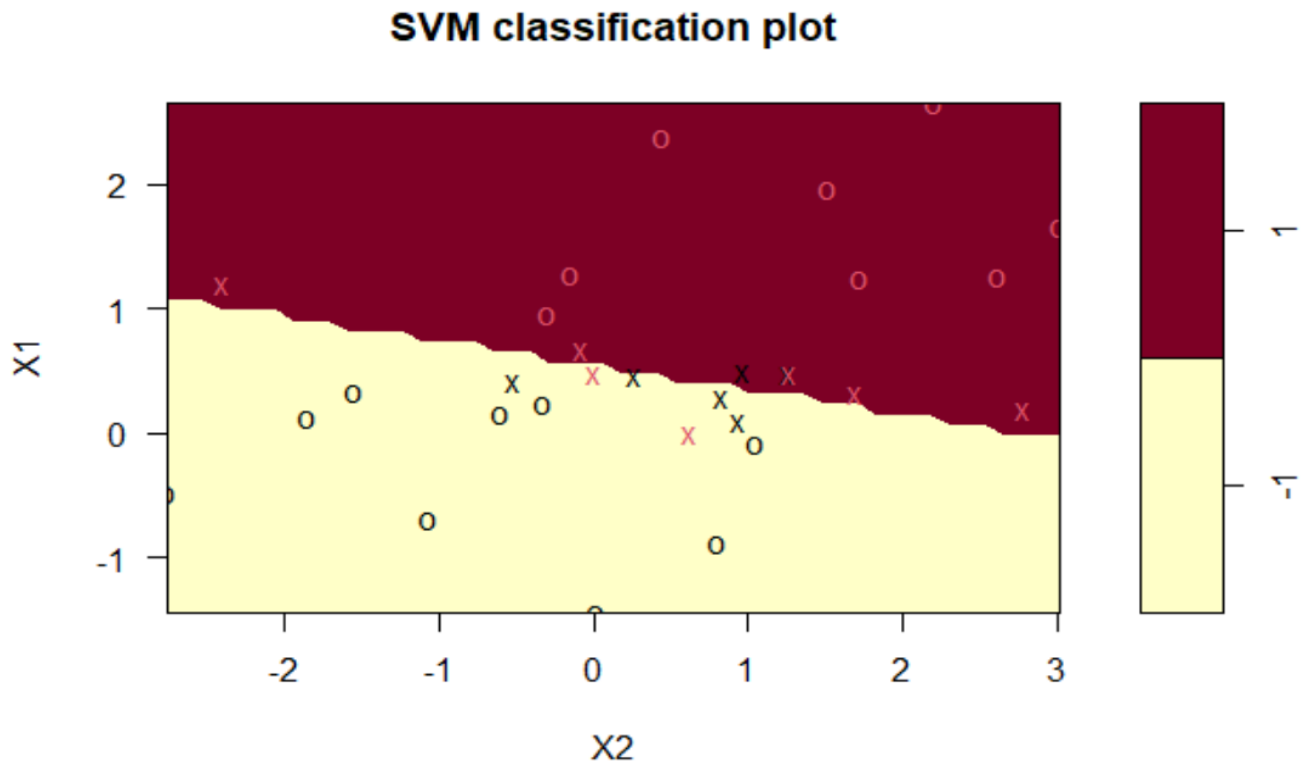
```
Number of Classes:  2
```

```
Levels:
```

```
-1 1
```

Inference: From the above output there were 13 support vectors, 6 in one class and 7 in another class.

`plot(svm,data)`



Inference: The support vectors as crosses and the remaining observations are plotted as circles.

2. Example: Write R code to find the number of support vectors for classifying the data. Data as follows: 100 observations, in two columns and also get the SVM classification plot.

R Code with Output:

```
x = matrix(rnorm(100), 50, 2)
y = rep(c(-1, 1), c(25, 25))
x[y == 1,] = x[y == 1,] + 1
data = data.frame(x, y = as.factor(y))
svm = svm(y ~ ., data = data, kernel = "linear", cost=10)
summary(svm)
```

```
call:
svm(formula = y ~ ., data = data, kernel = "linear",
     cost = 10)
```

```
Parameters:
  SVM-Type:  c-classification
  SVM-Kernel: linear
      cost:  10
```

```
Number of support vectors:  34
```

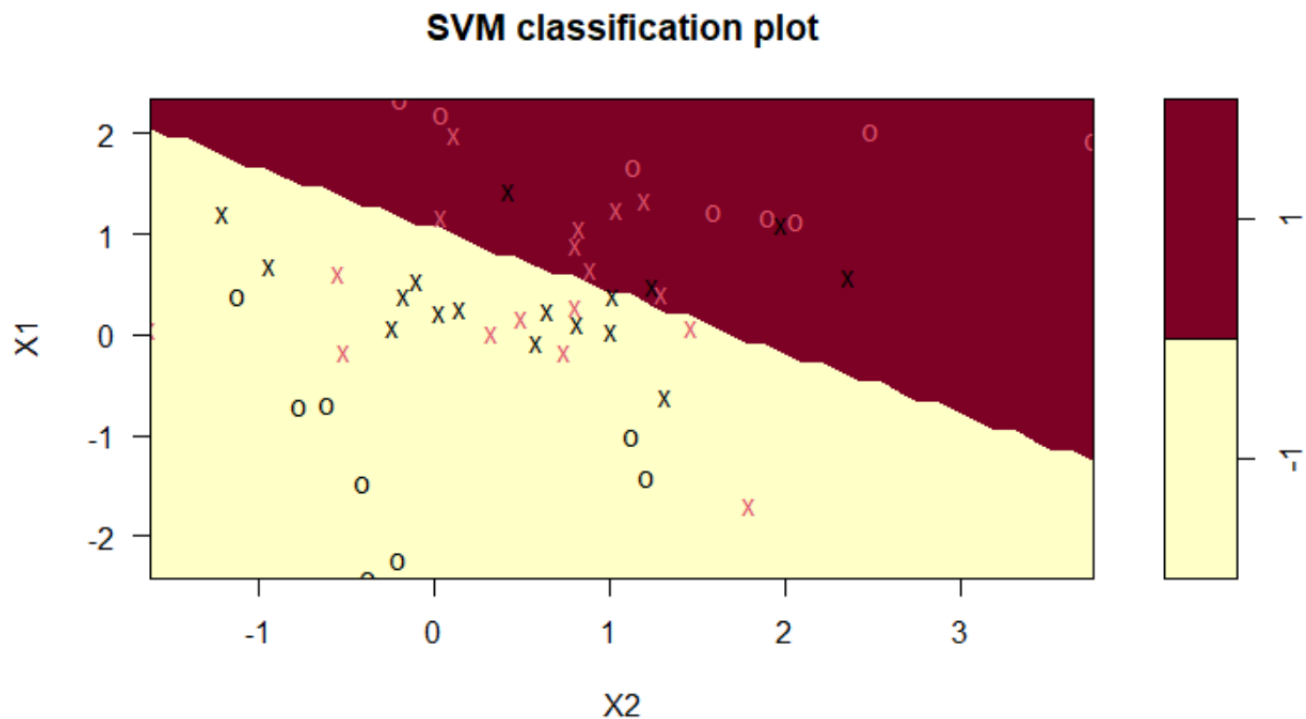
```
( 17 17 )
```

```
Number of classes:  2
```

```
Levels:
-1 1
```

Inference: From the above output there were 34 support vectors, 17 in one class and 17 in another class.

```
plot(svm,data)
```



Inference: The support vectors as crosses and the remaining observations are plotted as circles.

Viva Questions:

1 Define SVM?

Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems.

2. Define Hyperplane?

There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but we need to find out the best decision boundary that helps to classify the data points. This best boundary is known as the hyperplane of SVM.

3. Define support vectors?

The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector. Since these vectors support the hyperplane, hence called a Support vector.

9. Dimensionality reduction with PCA in R and K-means Clustering in R

9.1 Principal Component Analysis (PCA) in R

R Code:

Store the **Data** in Excel and Export the **Data**

PCA=prcomp(Data, scale=T)

summary(PCA)

1. Example: Obtain the percentage of principal components to the following data and interpret the results

X ₁	27	34	38	43	44	52	38	39	42	56
X ₂	65	74	70	68	63	60	59	62	61	65
X ₃	38	30	32	38	32	31	36	35	33	30
X ₄	67	65	66	54	63	62	61	60	56	55
X ₅	56	67	59	73	77	45	38	34	45	40

R Code with Output:

Store the **Data** in Excel and Export the **Data**

PCA=prcomp(PCA_DATA,scale=T)

summary(PCA)

```
> PCA=prcomp(PCA_DATA, scale=T)
> summary(PCA)
Importance of components:
              PC1      PC2      PC3      PC4      PC5
Standard deviation  1.4647  1.1965  0.9573  0.64346  0.3041
Proportion of Variance 0.4291  0.2863  0.1833  0.08281  0.0185
Cumulative Proportion 0.4291  0.7154  0.8987  0.98150  1.0000
```

Inference: PC1 explains 42% of the total variance implies half of information in the dataset and PC2 explains 28% of the total variance implies one-fourth of information in the dataset out of five variables can be encapsulated by one PC1. PC3 explains 18 % .

From the above components, one can observe that first variable corresponding to the X1 factor and second variable corresponding to the X2 factor can be included in the model.

2. Example: Obtain the percentage of principal components to the following data and interpret the results

X1	44	33	33	32	34	31	33	30	34	34	33	36	33	34	37
X2	1.3	2.2	2.2	2.6	2	1.8	2.2	3.6	1.6	1.5	2.2	1.7	2.2	1.3	2.6
X3	250	115	75	85	100	75	85	75	225	250	255	175	75	85	90
X4	0.63	0.59	0.56	0.55	0.54	0.59	0.56	0.46	0.63	0.6	0.63	0.58	0.55	0.57	0.62

R Code with Output:

PCA=prcomp(PCA_DATA, scale=T)

summary(PCA)

```
> PCA=prcomp(PCA_DATA, scale=T)
> summary(PCA)
Importance of components:

              PC1      PC2      PC3      PC4
Standard deviation   1.6436 0.7171 0.7055 0.53547
Proportion of Variance 0.6753 0.1285 0.1244 0.07168
Cumulative Proportion 0.6753 0.8039 0.9283 1.00000
```

Inference: PC1 explains 67% of the total variance implies two-third of information in the dataset out of four variables can be encapsulated by one PC1.

PC2 and PC3 explains 12 % each.

From the above components, one can observe that first variable corresponding to the X_1 factor can be included in the model.

9.2 K-means Clustering in R

Algorithm:

1. Install appropriate package
2. Store the **Data** in Excel and Export the **Data**
3. Choose k value by any one of the three methods
4. Apply kmeans clustering to the given data.

R Code:

```
install.packages("factoextra") # To visualize the clusters
library(factoextra)
```

To choose k value: 3 methods are (Elbow, Silhouette, Gap)

```
fviz_nbclust(Data, kmeans, method = "wss", k.max = 8) #k.max argument is optional
or
```

```
fviz_nbclust(Data, kmeans, method = "silhouette", k.max = 8)
```

or

```
fviz_nbclust(Data, kmeans, method = "gap", k.max = 8)
```

#From the above 3 methods k value will be taken

```
clust=kmeans(Data, k value)
```

```
clust
```

```
fviz_cluster(clust, Data) #Displays cluster plot
```

1 Example: The following is the set of 10 data points. Cluster these points using R code in k means algorithm.

S.No.	1	2	3	4	5	6	7	8	9	10
X ₁	18	22	25	32	38	42	44	46	52	60
X ₂	12	18	25	28	32	38	40	43	45	50
X ₃	18	22	35	38	39	43	50	55	60	65

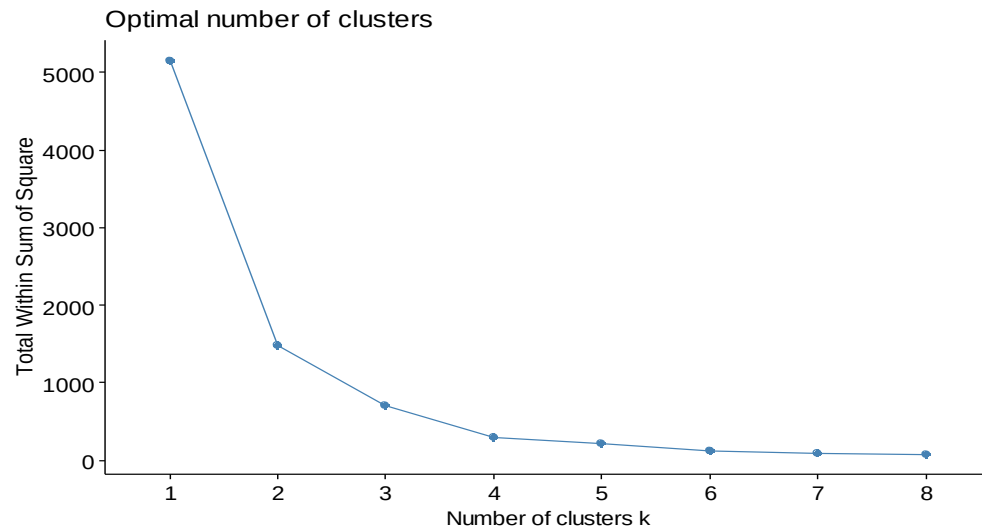
R Code with Output:

```
install.packages("factoextra")
```

```
library(factoextra)
```

```
fviz_nbclust(PCA_Data, kmeans, method = "wss", k.max = 8)
```

```
> library(factoextra)
> fviz_nbclust(PCA_Data, kmeans, method = "wss", k.max = 8)
```



Choose K=2

```
clust=kmeans(PCA_Data,2)
```

```
clust
```

```
> clust=kmeans(PCA_Data,2)
> clust
```

K-means clustering with 2 clusters of sizes 5, 5

Cluster means:

	X1	X2	X3
1	48.8	43.2	54.6
2	27.0	23.0	30.4

Clustering vector:

```
[1] 2 2 2 2 2 1 1 1 1 1
```

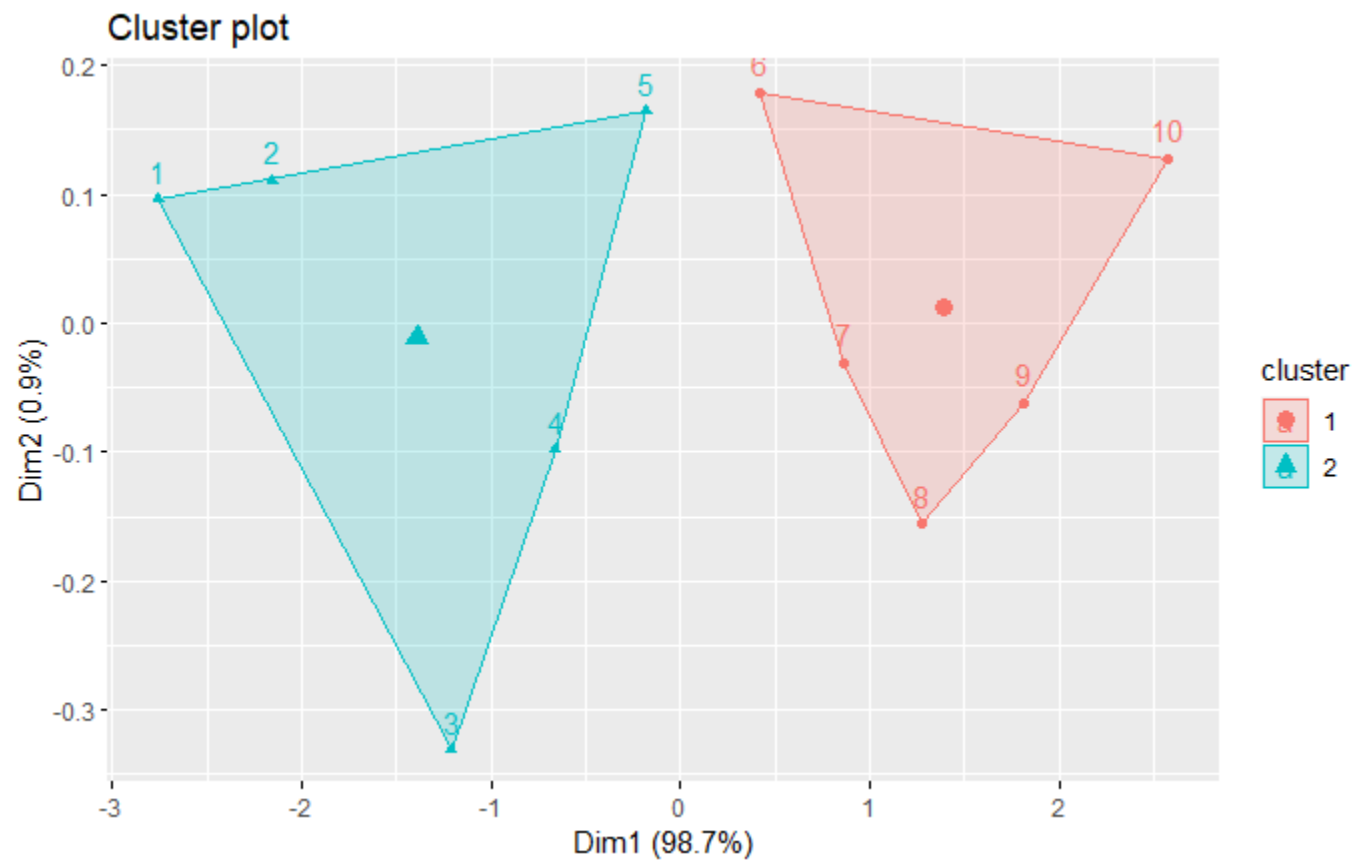
Within cluster sum of squares by cluster:

```
[1] 592.8 889.2
(between_SS / total_SS = 71.2 %)
```

Available components:

```
[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss" "betweenss"
[7] "size"         "iter"         "ifault"
```

```
fviz_cluster(clust,PCA_Data)
```



2 Example: Cluster these data points using R code in k means algorithm.

X1	44	33	33	32	34	31	33	30	34	34	33	36	33	34	37
X2	1.3	2.2	2.2	2.6	2	1.8	2.2	3.6	1.6	1.5	2.2	1.7	2.2	1.3	2.6
X3	250	115	75	85	100	75	85	75	225	250	255	175	75	85	90
X4	0.63	0.59	0.56	0.55	0.54	0.59	0.56	0.46	0.63	0.6	0.63	0.58	0.55	0.57	0.62

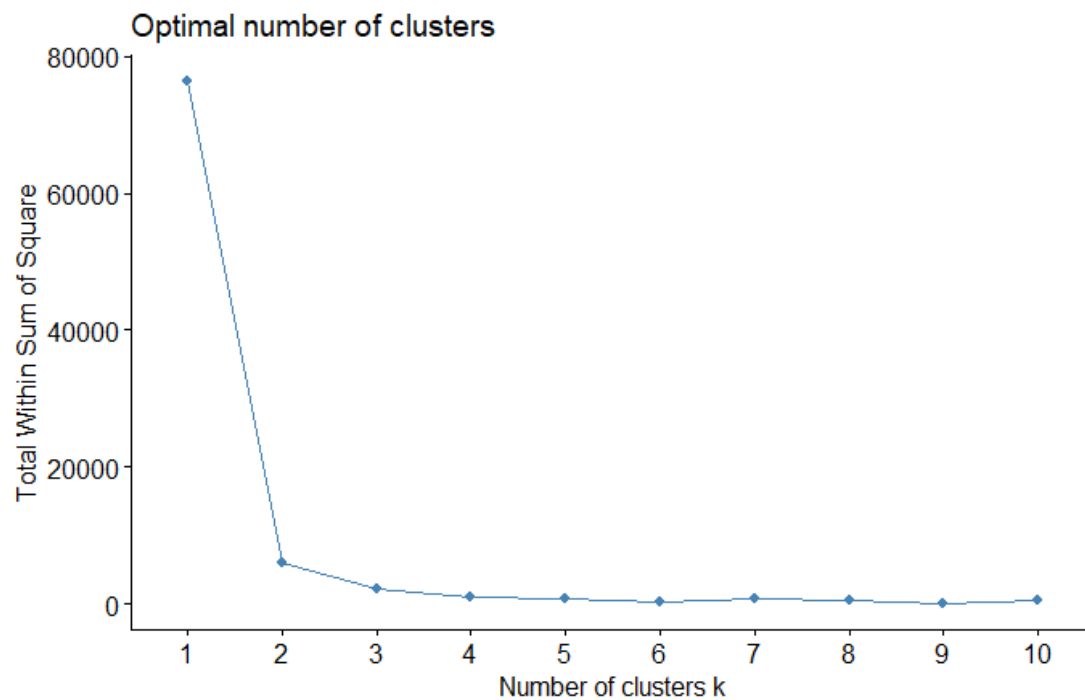
R Code with Output:

library(factoextra)

fviz_nbclust(PCA_DATA, kmeans, method = "wss", k.max = 10)

```
> library(factoextra)
```

```
> fviz_nbclust(PCA_DATA, kmeans, method = "wss", k.max = 10)
```



Choose k=2

clust=kmeans(PCA_DATA,2)

clust

```
> clust=kmeans(PCA_DATA,2)
> clust
K-means clustering with 2 clusters of sizes 5, 10
```

Cluster means:

	X1	X2	X3	X4
1	36.2	1.66	231	0.614
2	33.0	2.27	86	0.559

Clustering vector:

```
[1] 1 2 2 2 2 2 2 2 2 1 1 1 2 2 2
```

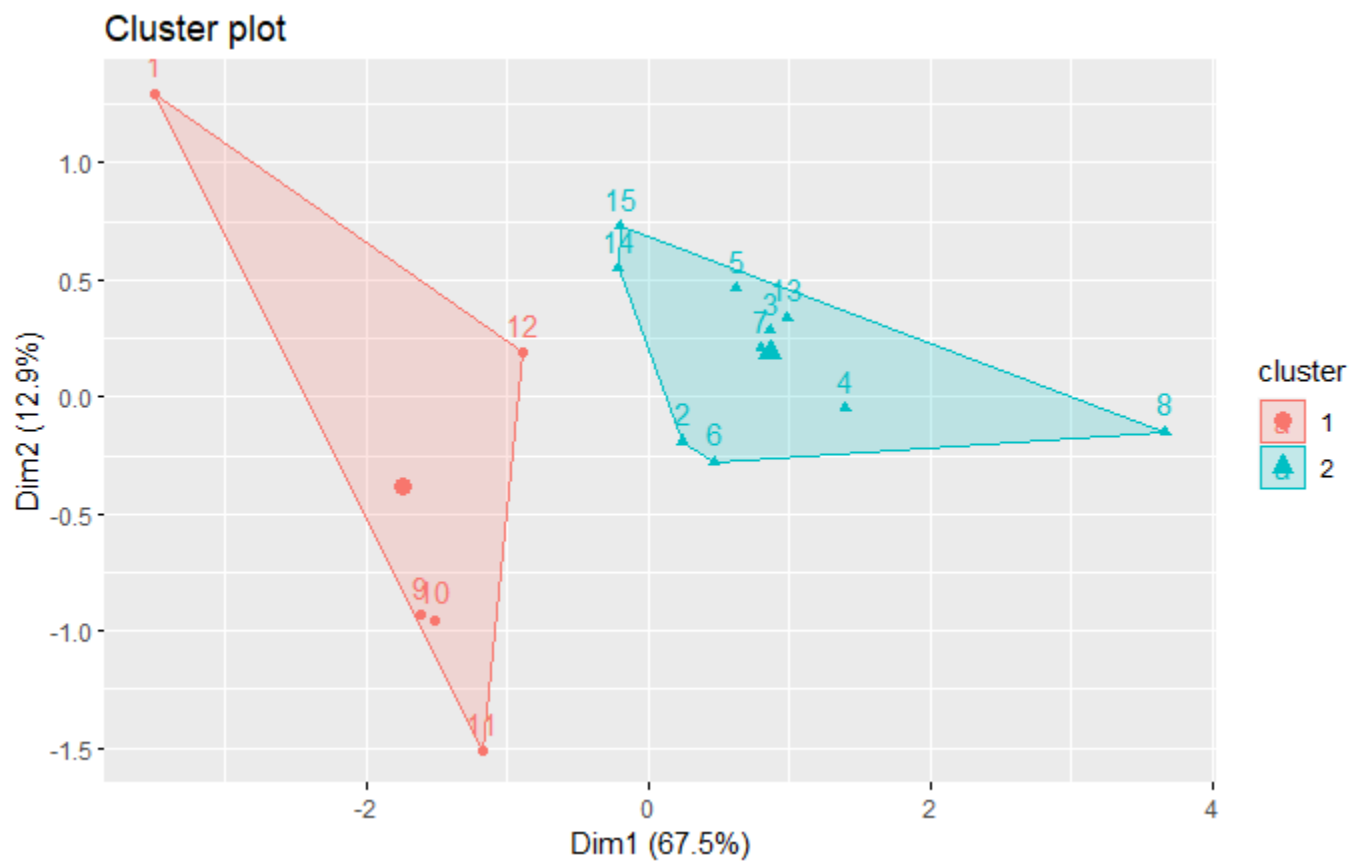
Within cluster sum of squares by cluster:

```
[1] 4551.254 1575.257
(between_SS / total_SS = 92.0 %)
```

Available components:

[1] "cluster"	"centers"	"totss"	"withinss"	"tot.withinss"	"betweenss"
[7] "size"	"iter"	"ifault"			

fviz_cluster(clust,PCA_DATA)



Viva Questions:

1. Define Cluster.

Cluster is defined as collections or groups of items with similar or different characteristics.

2. Define Principal Components.

Principal component analysis (PCA) is a statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables (entities each of which takes on various numerical values) into a set of values of linearly uncorrelated variables called principal components.

3. What kind of Machine learning did PCA and Clustering techniques?

PCA and Clustering techniques belongs to Un supervised Machine Learning

10. Kernel SVM in R

Kernel SVM:

R Code:

```
install.packages("e1071")
library(e1071)
x = matrix(rnorm(32), 16, 2)      #Generating the data
y = rep(c(-1, 1), c(8, 8))      #Assigning +1 and -1 to the above data
x[y == 1,] = x[y == 1,] + 1     #bias the x variable when y =1
dat = data.frame(x, y = as.factor(y)) #representing the above data in table form using dataframe
svm = svm(y ~ ., data = dat, kernel = "radial", gamma=1, cost=1)
summary(svm)
plot(svm, dat)
```

1 Example: Write R code to find the number of support vectors for classifying the data in radial basis kernel. Data as follows: 400 observations, in two columns and also get the SVM classification plot.

R Code with Output:

```
x = matrix(rnorm(400), 200, 2)
y = rep(c(-1, 1), c(100, 100))
x[y == 1,] = x[y == 1,] + 1
data = data.frame(x, y = as.factor(y))
svm = svm(y ~ ., data = data, kernel = "radial", gamma=1, cost=1)
summary(svm)
```

```
Call:
svm(formula = y ~ ., data = data, kernel = "radial", gamma = 1,
    cost = 1)

Parameters:
  SVM-Type:  C-classification
 SVM-kernel: radial
      cost:  1

Number of Support Vectors:  134

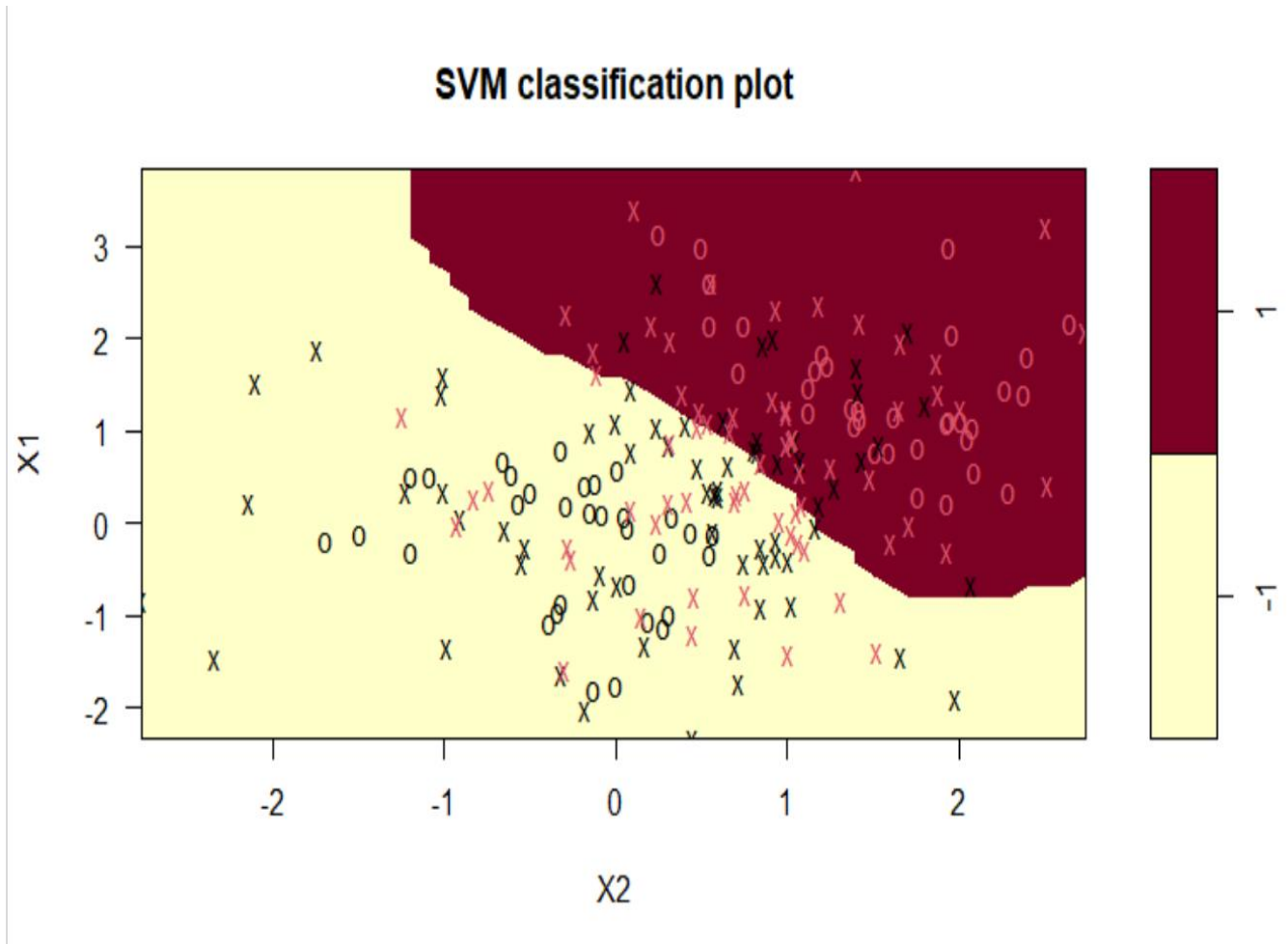
( 68 66 )

Number of Classes:  2

Levels:
-1 1
```

Inference: From the above output there were 134 support vectors, 68 in one class and 66 in another class.

`plot(svm,data)`



Inference: The support vectors as crosses and the remaining observations are plotted as circles.

2 Example: Write R code to find the number of support vectors for classifying the data in radial basis kernel. Data as follows: 300 observations, in two columns and also get the SVM classification plot.

R Code with Output:

```
x = matrix(rnorm(300), 150, 2)
```

```
y = rep(c(-1, 1), c(75, 75))
```

```
x[y == 1,] = x[y == 1,] + 1
```

```
data = data.frame(x, y = as.factor(y))
```

```
svm = svm(y ~ ., data = data, kernel = "radial", gamma=1, cost=10)
```

```
summary(svm)
```

```
call:
svm(formula = y ~ ., data = data, kernel = "radial", gamma = 1, cost = 10)
```

```
Parameters:
  SVM-Type:  C-classification
  SVM-Kernel: radial
    cost:   10
```

```
Number of Support Vectors: 91
```

```
( 43 48 )
```

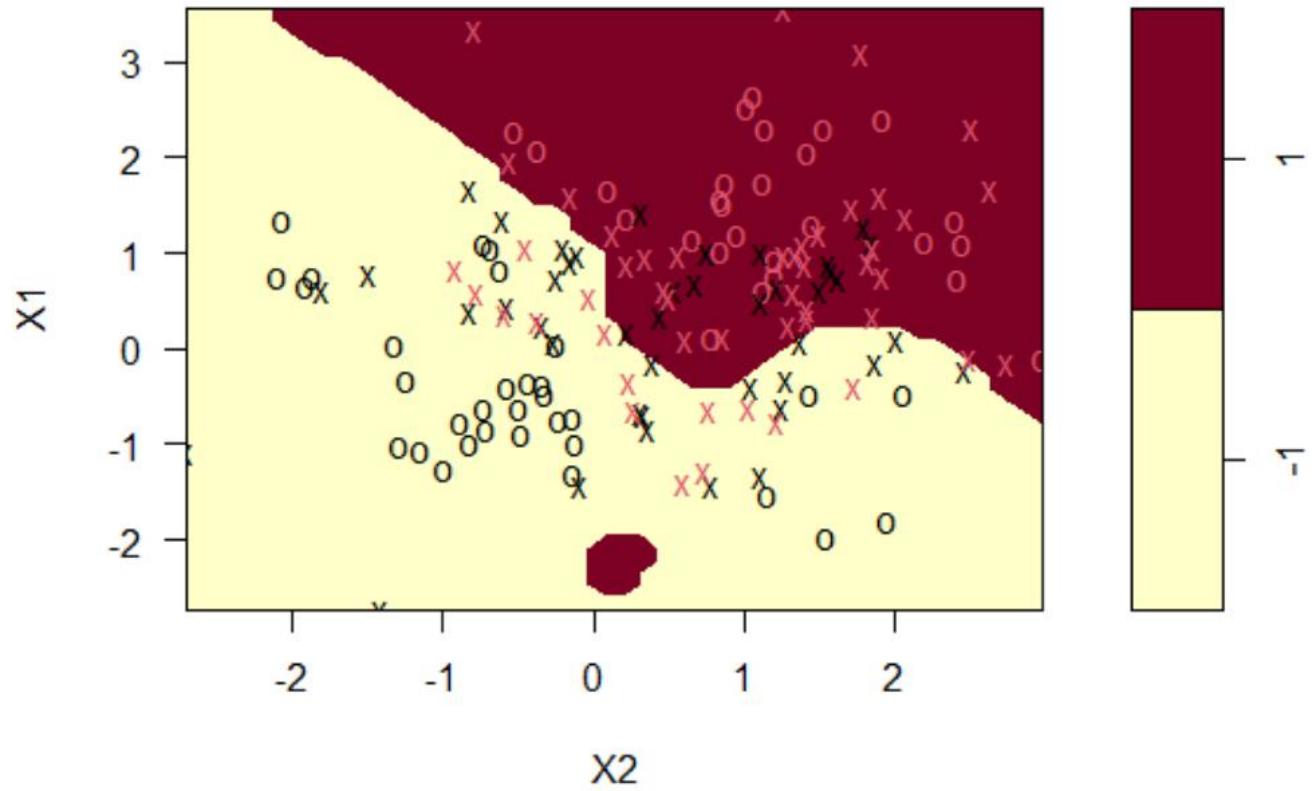
```
Number of Classes: 2
```

```
Levels:
-1 1
```

Inference: From the above output there were 91 support vectors, 43 in one class and 48 in another class.

```
plot(svm,data)
```

SVM classification plot



Inference: The support vectors as crosses and the remaining observations are plotted as circles.

Viva Questions:

1. Define Kernel.

Kernel Function is a method used to take data as input and transform it into the required form of processing data. “Kernel” is used due to a set of mathematical functions used in Support vector Machine providing the window to manipulate the data.

2. What is the use of Kernel

In machine learning, a “kernel” is usually used to refer to the kernel trick, a method of using a linear classifier to solve a non-linear problem.

List of Additional programs**1. Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R programming****(i) Uniform Distribution****R Code:****runif(n, min, max)****1. Example:** Generate 5 Uniform random numbers from 10 to 100.**R Code with Output:****runif(5,10,100)**

```
[1] 42.61054 57.56261 23.57929 76.08863 25.25279
```

2. Example: Generate 10 random samples from the Uniform Distribution $U(0,1)$ **R Code with Output:****runif(10,0,1)**

```
[1] 0.67139102 0.34191665 0.50767938 0.89660762 0.05948337 0.65363865
[7] 0.83952941 0.12345681 0.97254724 0.75465741
```

(ii) Normal Distribution**R Code:****rnorm(n, mean, sd)****1. Example:** Generate four random numbers from $N(70, 10)$ **R Code with Output:****rnorm(4,70,sqrt(10))**

```
[1] 74.84571 70.68275 67.49411 67.07436
```

2. Example: Generate 5 Random samples from the Normal distribution $N(25,4)$

R Code with Output:

```
rnorm(5,25,sqrt(4))
```

```
[1] 24.53206 24.90118 29.06580 23.51796 23.38991
```

(iii) Binomial Distribution

R Code:

```
rbinom(no of random numbers to be generate, n, p)
```

1. Example: Generate 6 Random samples from the Binomial distribution $B(10, 0.5)$

R Code with Output:

```
rbinom(6,10,0.5)
```

```
[1] 4 5 6 6 5 5
```

(iv) Exponential Distribution

R Code:

```
rexp(n, theta )
```

1. Example: Generate 5 Random samples from the Exponential distribution with parameter 10

R Code with Output:

```
rexp(5,10)
```

```
[1] 0.007909765 0.021077747 0.182569774 0.156448570 0.125258499
```

2. Multiple linear regression

Testing overall hypothesis and testing significance of individual variables, model selections and prediction in R programming is called Multiple linear regression.

R Code:

`lm(y ~ x1+x2+x3...,data)`

Example 1: Fit a multiple linear regression to the following data set.

S.No.	1	2	3	4	5	6	7	8	9	10	11	12
X1	70	77	72	77	74	80	82	74	72	77	90	82
X2	100	105	112	105	125	112	129	125	112	105	140	129
X3	125	110	118	110	122	123	121	122	118	110	128	121
X4	65	70	66	70	58	68	61	58	66	70	82	61
Y	77	73	68	76	73	77	80	73	72	74	66	78

R Code with Output:

Left for student exercise.

Programme Outcomes

S N o.	CO	Name of the Programme	Outcome of the Programme	Mention any Real-time application where we can apply this method	Which one is more efficient
1	CO1	Introduction to R Programming	Perform built-in commands and operations with matrices in R-programming. Analyze the appropriate data type of variables and evaluate various Frequency Distributions in R - programming. Determine the mean and standard deviation of a sampling distribution using R.	R is a open source Software and it can be easily accessed. It represents the whole set of data by a single value. and gives us the location of central points.	
		Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)			
		Acceptance/Rejection Sampling in R			
		Maximum likelihood and method of moments estimation.			
2	CO2	Testing of Hypothesis based on Z-test. Concept of p-value in R. Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R	Develop a code for testing of hypothesis using Z-test, t-test, chi-square-test, F-test.	To test the significance of hypothesis based on the p-value	
3	CO3	Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R	Classify Linear and Logistic regression.	To classify the objects into groups	
		Logistic Regression in R			
4	CO4	Classification with SVM in R	Write Code for SVM and Kernel SVM methods.	To classify the objects into groups when the data is large.	
		Dimensionality reduction with PCA in R and K-means Clustering in R			
		Kernel SVM in R			



...striving towards perfection